Tail-Risk Pricing and Collective AI Safety: Insurance under Dependence Ambiguity

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Abstract

Can greater concern about catastrophic AI losses weaken incentives for collective safety? We study insurance with known marginal loss frequencies and ambiguous dependence. Increasing the weight on a fixed tail statistic raises savings per insured exposure but contracts coverage; assurance value can decline while both counterfactual markets remain active. Focusing on a more extreme tail creates a distinct effect: a control that leaves the same largest payout credible has vanishing insurance-market value. We then make physical control costly and endogenous. In the common-regime model, positive residual risk at every finite effort and unboundedly increasing cost make zero investment optimal under sufficiently extreme tail focus, despite active insurance. With exponential dependence reduction, optimal positive effort increases as the assessed tail narrows, then drops discontinuously to zero. Wider dependence ambiguity can produce the same intensification followed by abandonment. Finite-cost elimination of common risk supplies a countercase, and finite-pool and extreme-tail limits need not commute. Private benefits add a separate collection constraint. The analysis distinguishes generated market value, its distribution, and its collectibility; it does not measure the total social value of AI safety.

Keywords: AI insurance; Knightian uncertainty; dependence ambiguity; tail risk; expected shortfall; collective assurance.
JEL codes: D81, D62, G22, G28.

1 Introduction

An insurance market for artificial intelligence must price more than the error rate of a single application. Firms may share an underlying model, a software dependency, a cloud service, or a model update. A failure can therefore generate simultaneous claims from economically distinct policyholders. Novel systems also create an epistemic problem: neither the probability of joint failure nor the set of relevant failure modes need be well established. These are separate problems. Dependence concerns the joint loss law; ambiguity concerns the credibility of that law.

This paper asks when a market can form if individual loss frequencies are known but dependence remains ambiguous. Holding the marginal distribution fixed isolates a mechanism that improvements in average accuracy cannot identify. An insurer can face the same expected loss per policy under every admissible model and nevertheless quote very different prices as its set of credible dependence models changes.

We analyze a simple common-regime environment. In one regime, losses are idiosyncratic and average out in a large pool. In the other, a shared event determines all losses. The probability of entering the

common regime is unknown within a specified interval. Insurers evaluate their bounded liabilities through a mean–expected-shortfall certainty equivalent and take the worst case over that interval. Customers have heterogeneous willingness to pay for a fixed insurance layer. Competitive prices compensate the insurer for its specified risk-bearing preferences; they are neither actuarially fair expected-loss prices nor statutory capital requirements.

The main mechanism has two thresholds. A sufficiently pessimistic dependence bound makes the upper tail charge saturate. Small improvements then leave the price unchanged. Even after the price begins to fall, it may remain above customers' willingness to pay. These two thresholds can make a large assurance intervention valuable while small interventions earn no insurance-market return. With a linear assurance technology and linear investment cost, the resulting gains from trade are convex in investment. The optimum is consequently at an endpoint. Shared benefits then permit a no-investment equilibrium despite positive collective gains.

The central economic result is a nonmonotone return to assurance. Greater tail sensitivity raises the price saving on each insured exposure but reduces the quantity of coverage sold. We first show the mechanism with discrete participation, then allow continuously varying demand and convex costs of supplying funded capacity. If $Q(a)$ is equilibrium coverage at tail surcharge a, and the project reduces the per-unit tail loading from d_0 to d_1, its gross market value is $\int_{\lambda d_1}^{\lambda d_0} Q(a)\,da$. The sign of its response to tail-risk weight λ is governed by $d_0 Q(\lambda d_0) - d_1 Q(\lambda d_1)$. The response can be negative while both markets remain active. Under linear demand the peak occurs strictly before the original market closes.

Tail weight and tail focus are different experiments. Increasing the ES confidence level changes the losses the pricing criterion distinguishes. For arbitrary sets of bounded loss laws with a common mean, a control that leaves the same worst admitted payout has vanishing market value in the extreme tail. This can occur while both insurance markets remain active. In the common-regime benchmark, we derive the residual dependence bound required for market gains to cover a fixed project cost: it must decline in proportion to the assessed tail probability. This condition concerns a credible bound delivered by the control, not permission to discard adverse models. A finite-pool audit also shows why the large-pool and extreme-tail limits cannot always be interchanged.

Endogenous physical control turns this funding requirement into an investment choice. Suppose some common risk remains at every finite effort and resource cost tends to infinity. Market gains have a finite upper bound, so sufficiently extreme tail pricing makes all potentially affordable controls ineffective at lowering the premium. Zero investment then becomes optimal while coverage remains available. With exponential dependence reduction and linear effort cost, we solve the global choice: positive effort rises as the assessed tail narrows, preserving a constant improved surcharge, until cost absorbs the available gain and the project is abandoned. A wider admitted dependence bound can create the same pattern at a fixed true initial risk. Complete removal of common risk at a sufficiently low finite cost is a countercase. These are collective market-surplus choices, not a decentralized investment equilibrium.

Ambiguity has a separate comparative static. Holding the true loss process and the physical intervention fixed, a wider set of plausible dependence parameters increases underwriting prices. It need not increase the intervention's market value: the wider set also makes residual risk after the intervention more expensive. This separates uncertainty about risk from an actual deterioration in risk. Neither result means that a safer system has less social value. A contraction of insurance can reduce financeable market gains while leaving substantial tail exposure with policyholders.

A second constraint separates a valuable project from a financeable mechanism. When benefits are public, sequential commitments and credible cancellation can finance a beneficial intervention. With private benefits, we instead require truthful reporting in dominant strategies, participation relative to no project, and efficient project selection at every type profile. A standard critical-payment argument yields an exact uniform funding bound. Tail pricing can raise this bound while increasing collective gains at the same profile of customers. The reason is that participation in insurance determines how much of each customer's benefit is publicly guaranteed. Allowing one contract to allocate insurance and collect project payments does not relax the bound if both allocations remain efficient. Verifiable information about earlier purchases can matter, but changing the form of a contract alone does not supply that information. A self-financing unanimity rule remains possible, but can reject projects with positive aggregate gains. The result is a tradeoff among information, efficiency, and funding, not a claim that every valuable project needs a subsidy.

Two extensions examine how the gains are distributed. With fixed provider fees, wider dependence ambiguity weakens the diversification credit and can concentrate adoption, whereas greater weight on a fixed tail statistic strengthens that credit. When model providers choose prices, the incidence depends on portfolio-cost curvature and risk-bearing supply. A safety improvement can lower insurance quotes while higher vendor fees reduce customer surplus. A fixed endowment of risk tolerance preserves this possibility in a weak-risk domain, but introduces insurer rents that must enter the accounting. These qualifications are developed in the appendices; the main contribution concerns the value and funding of collective controls under ambiguous tails.

The argument concerns a specified, bounded contract. A conservative ambiguity set can be appropriate when deployment data are sparse or a shared component changes. Nonmonotone prevention incentives, public-project payment constraints, and ambiguity-induced concentration all have established precedents. The application here connects these forces to observable insurance prices, collective safety finance, and the architecture generating common losses. The fixed-fee architecture allocation is efficient under its maintained criterion. The provider-pricing extension separately introduces markups and audits their effect; neither benchmark establishes a general welfare case against prudent underwriting.

2 Relation to existing research

Three established literatures constrain the contribution. Multiple-prior decision theory supplies a disciplined alternative to assigning a single trusted probability law (Gilboa and Schmeidler, 1989). Catastrophe-insurance research already links ambiguous scientific models to insurance prices and availability; Dietz and Niehörster (2021) study ambiguity pricing using hurricane models. Robust actuarial analysis also treats uncertain dependence explicitly (Koike, 2026). Merely replacing a point estimate with a worst-case expected shortfall is therefore not a new contribution.

AI insurance research already addresses underwriting, incentives, and shared exposures. Ge and Zhu (2023) study liability insurance in an AI diagnosis setting. Szpruch et al. (2026) develop an insurance-based approach to AI deployment incentives. Zhu (2026) discuss actuarial exposures and accumulation for agentic systems. The common-shock expected-shortfall floor in Nagaraj and Lee (2026, Section 2.4.1) is particularly close to any claim based only on non-diversification. Our elementary tail calculation is an input, not a claim to have discovered systemic AI risk.

Laszka and Grossklags (2015) already model a monopolistic cyber-insurer investing in widely used software

to reduce shared losses, with explicit insurance demand and safety capital. Thus insurer-funded common technical protection predates AI insurance. Their investment changes software vulnerability and individual incident probabilities. Our main control holds marginal losses fixed and isolates the pricing and financing consequences of changing dependence.

The closest overlap for the comparative static is Li et al. (2025, Sections 5.1.1–5.1.2), whose numerical examples show prevention effort first rising and then falling with premium loading as insurance demand contracts. Their control changes an individual's loss probability. Here marginal losses are fixed, the control changes shared dependence, and we characterize collective market gains under ambiguity. Muermann and Kunreuther (2008) analyze underinvestment in self-protection with risk interdependencies. Our financing analysis builds on that established connection between protection, insurance, and externalities.

Alary et al. (2013, Section 4) give conditions under which greater ambiguity aversion lowers self-protection. Our costly-control comparison instead widens an underwriting ambiguity set at a fixed evaluation criterion and holds marginal claim probabilities constant. It is not a general result about ambiguity aversion or a claim that adverse prevention responses to uncertainty are new.

Shared assurance also connects to the standard problem of public-good provision (Samuelson, 1954; Bergstrom et al., 1986). The private-benefit analysis uses the critical-payment logic of Groves (1973) and Green and Laffont (1977). The general tension between efficient provision, incentive compatibility, and financing is established territory. Our application makes the benefit domain endogenous to tail-risk pricing and computes the associated funding requirement.

Trout (2025, Sections 2–3) proposes mandatory government AI indemnification, expert risk elicitation, and insurance-revenue support for shared safety research. This is direct prior art for linking AI insurance and collective safety funding. Our model studies voluntary insurance demand and a prior-free private-benefit benchmark with explicit payment constraints. Its funding bound does not evaluate that proposal's mandatory fees, risk-survey mechanism, or quadratic financing.

Jiang et al. (2022) show that ambiguity about return correlations can produce portfolio concentration under known marginals and multiple-prior evaluation. The architecture extension uses that established force in a market for insured technology choices. Its specific objects are the competitive premium difference, its expected-shortfall saturation threshold, and aggregate claim exposure under a fixed true law. We do not claim that ambiguous dependence first becomes a reason to concentrate in this setting.

The tail-focus analysis uses established ES and spectral-risk properties (Acerbi and Tasche, 2002; Acerbi, 2002). It does not claim a new endpoint theorem for risk measures. Its economic objects are the market value of a dependence control, the residual bound required to cover its cost, and the limits of a large-pool approximation. Li et al. (2025, Section 5.1.3) also vary an individual's TVaR confidence level and find nonmonotone self-protection. A broad claim that greater tail concern can reduce prevention is therefore insufficient for novelty.

The pricing extensions likewise use established methods. Global deviation checks are standard in differentiated-product competition (d'Aspremont et al., 1979); negative network effects can soften competition (Thisse et al., 1995); and reducing adverse selection can worsen insurance prices and coverage under market power (Mahoney and Weyl, 2017). Here marginal claim laws are identical and strategic prices belong to model providers. Utility-based equilibrium premiums and aggregation of exponential risk tolerances are established by Bühlmann (1980, Sections 2–6). The appendix comparison applies these methods to the incidence of a dependence control and makes insurer rents explicit.

3 Environment and actuarial benchmark

3.1 A fixed contract and a dependence ambiguity set

Time consists of an assurance decision, insurance contracting, and a single loss realization. The insured layer pays one unit for a qualifying loss and zero otherwise. It is a monetary contract limit, not an assertion that the underlying social harm is bounded by one. Fix a marginal claim probability $p \in (0, 1)$.

For a finite pool of n exposures, let $R \sim \text{Bernoulli}(\theta)$ select the regime. Conditional on $R = 1$, every exposure has the same loss $B \sim \text{Bernoulli}(p)$. Conditional on $R = 0$, losses B_i are independent $\text{Bernoulli}(p)$. All these draws are independent before selecting the regime. Then

$$L_i = RB + (1 - R)B_i, \qquad \mathbb{E}L_i = p, \qquad \text{Corr}(L_i, L_j) = \theta \quad (i \neq j). \tag{1}$$

The equality between θ and correlation is special to this family. An ordinary correlation estimate does not characterize general joint-tail risk.

The large-pool average has the distribution

$$Y_\theta = \begin{cases} 1 & \text{with probability } \theta p, \\ p & \text{with probability } 1 - \theta, \\ 0 & \text{with probability } \theta(1 - p). \end{cases} \tag{2}$$

Its mean is always p. The underwriting set is

$$\mathcal{P}(u) = \{\mathcal{L}(Y_\theta) : 0 \leq \theta \leq u\}, \qquad u \in [0, 1]. \tag{3}$$

No probability distribution over θ is imposed. This is a restricted representation of Knightian uncertainty about dependence, not a representation of all unknown unknowns. The state space, marginal frequency, and dependence family are maintained modeling assumptions.

3.2 Expected shortfall and saturation

Let $\alpha \in (0, 1)$ and $\tau = 1 - \alpha$. We use the tail-average definition valid at atoms,

$$\text{ES}_\alpha^P(Z) = \inf_{t \in \mathbb{R}} \left\{ t + \frac{1}{\tau}\mathbb{E}_P(Z - t)_+ \right\}, \qquad r(u) = \sup_{P \in \mathcal{P}(u)} \text{ES}_\alpha^P(Y). \tag{4}$$

This is the standard optimization representation (Rockafellar and Uryasev, 2002). It avoids equating expected shortfall with a conditional expectation above a quantile when the distribution has atoms.

Lemma 1 (Tail charge; an elementary benchmark)**.** *The robust charge and its saturation threshold are*

$$r(u) = \min\left\{1, \frac{p}{\tau}, p + \frac{up(1 - p)}{\tau}\right\}, \tag{5}$$

$$u_* = \min\left\{\frac{\tau}{p}, \frac{\alpha}{1 - p}\right\}. \tag{6}$$

It increases linearly below u_* and is constant above u_*. The worst case is attained at $\theta = u$.

Proof. The upper τ-tail first includes mass at one, next at p, and finally at zero. If $\theta p \geq \tau$, its mean is one. If $\theta p < \tau \leq \theta p + 1 - \theta$, the tail mean is $p + \theta p(1-p)/\tau$. Otherwise all positive losses fit inside the tail, and its mean is p/τ. The three expressions agree at the boundaries and give (5). Monotonicity gives the worst-case endpoint and (6). □

For $p \geq \tau$, saturation occurs because the common-loss state alone fills the tail. For $p < \tau$, it occurs when the tail contains all positive losses and enough zeros. Neither result requires an unbounded heavy-tailed severity distribution. Here "tail risk" means large aggregate payouts relative to normal aggregate losses. Heavy-tailed social losses are a different extension.

3.3 Insurance demand, risk-bearing preferences, and payment

There are N client associations, each representing one unit of exposure mass and a sufficiently large population of small clients for (2) to be the relevant aggregation benchmark. Association i has a fixed per-unit monetary willingness to pay $w_i \in [p, 1]$ for full coverage of the specified layer. The association evaluates the sum of its clients' monetary certainty-equivalent gains. For an individual client with constant absolute risk aversion $a > 0$ and binary unit loss,

$$w(a, p) = a^{-1} \log(1 - p + pe^a). \tag{7}$$

This motivates dependence-invariant marginal willingness to pay; it is not a utility representation for an association owning the entire correlated portfolio itself.

Competitive insurers have the monetary risk-bearing valuation

$$\rho_u(Z) = (1 - \lambda)\mathbb{E}Z + \lambda \sup_{P \in \mathcal{P}(u)} \mathrm{ES}_\alpha^P(Z), \qquad 0 < \lambda \leq 1, \tag{8}$$

on liabilities whose mean is common across the admitted laws. They require premium revenue at least $\rho_u(Z)$. This is a specified coherent, risk-sensitive participation rule. It is not derived from a representative-agent consumption model. For a quota of exposure mass x, $Z = xY$, so competition gives

$$c(u) = p + \lambda[r(u) - p], \qquad b_i(u) = [w_i - c(u)]_+. \tag{9}$$

Competitive profit is zero in this risk-adjusted sense. Expected accounting profit can be positive as compensation for bearing risk.

The ambiguity contribution can be separated from the underlying common risk. If the true dependence θ were known, the same criterion would quote $c(\theta)$. The ambiguity wedge is

$$c(u) - c(\theta) = \lambda[r(u) - r(\theta)] \geq 0 \quad \text{for } \theta \leq u. \tag{10}$$

It can prevent trade when $c(\theta) < w_i < c(u)$, despite identical marginal loss probabilities. It can also be zero when both bounds lie in the saturated region. Thus ambiguity is not synonymous with correlation, and nonzero ambiguity need not change every underwriting decision. The threshold and investment mechanisms can also arise under known dependence; uncertainty changes their location rather than being necessary for their existence.

Insurers are assumed able to pay the maximum liability of the layer from external prefunded assets. This removes default from the pricing mechanism. The number $r(u)$ is a risk charge, not available cash, an equity amount, or a guarantee of payment. The initial benchmark normalizes capacity costs and tax to zero. Section 5 introduces costly funded capacity and endogenous coverage; Section 8 consolidates payment resources. Neither section solves an endogenous default equilibrium.

4 Assurance investment and market formation

4.1 A verifiable change in the admitted dependence family

Let $e \in [0,1]$ index a shared assurance intervention with cost Fe, where $F > 0$. It delivers the verified bound

$$u(e) = u_0 - de, \qquad d = u_0 - u_1 > 0, \qquad 0 \le u_1 < u_0 \le 1. \tag{11}$$

The intervention is a *verified control technology*. One implementation starts with an unknown common-regime probability $q \in [0, u_0]$ and replaces a fraction $(d/u_0)e$ of common regimes with independent regimes, retaining each marginal p. The resulting dependence is $\theta(e) = q(1 - de/u_0)$. A trusted audit verifies delivery of the control. The admitted bound then follows mechanically.

This assumption is strong but coherent: the intervention changes the system. A pure diagnostic test cannot be assumed to produce favorable evidence with certainty when the true dependence is unknown. Section 9 treats that distinction. In an AI application, the linear attenuation rule would require independent validation; model diversity alone does not establish it.

Define the aggregate incremental money-metric gains from trade as

$$B(e) = \sum_{i=1}^{N} \{[w_i - c(u(e))]_+ - [w_i - c(u_0)]_+\}. \tag{12}$$

This sums client certainty-equivalent gains, with insurers compensated at their risk-sensitive reservation prices. It is a stated risk-adjusted welfare criterion; the risk charge is not a physical destruction of resources, and this is not a general utilitarian welfare theorem.

Proposition 1 (Threshold and all-or-nothing assurance)**.** *Suppose* (11) *holds.*

1. *If $u_0 > u_*$, the insurance price is unchanged for all e such that $u(e) \ge u_*$. In particular, the initial price response is zero.*
2. *$B(e)$ is nondecreasing and convex on $[0,1]$. Therefore a maximizer of $B(e) - Fe$ exists at an endpoint. If $B(1) > F$, full assurance is the unique optimum; if $B(1) < F$, zero is the unique optimum.*
3. *For identical w with $p < w < c(u_0)$ and $c(u_1) < w$, market participation starts only after $u(e)$ falls below*

$$u_w = \frac{(w-p)\tau}{\lambda p(1-p)} < u_*. \tag{13}$$

The entry threshold is $e_w = (u_0 - u_w)/d$.

Proof. The first claim follows from Lemma 1. Write $A = p(1-p)/\tau$ and $M = \min\{1, p/\tau\}$. Then $r(u(e)) = \min\{M, p + A(u_0 - de)\}$ is concave and nonincreasing in e. Thus $w_i - c(u(e))$ is convex and nondecreasing, as is its positive part. Summing gives the claim for B. Since $B(0) = 0$, convexity implies $B(e) - Fe \le e[B(1) - F]$ for $0 \le e \le 1$. The endpoint conclusions follow. Below saturation, $c(u) = p + \lambda A u$; solving $c(u) = w$ gives the entry threshold. The assumed willingness to pay places it strictly below saturation and strictly between u_1 and u_0. □

If $B(1) = F$, both endpoints are optimal and intermediate choices can tie. Convex investment costs can produce an interior optimum. Also, a nonzero initial price response is possible when $u_0 < u_*$; the threshold is a conditional prediction, not a universal property of uncertainty.

4.2 Shared benefits and the financing equilibrium

To make the financing game explicit, each association chooses a nonnegative contribution g_i. Investment is $e = \min\{\sum_i g_i/F, 1\}$. Each contributor pays its contribution, including any excess above the provision cost. Its incremental payoff is

$$\beta_i(e) - g_i, \qquad \beta_i(e) = [w_i - c(u(e))]_+ - [w_i - c(u_0)]_+. \tag{14}$$

Benefits are public to the client group even if a particular association contributes nothing; competitive insurers can use the verified system improvement.

Proposition 2 (A no-assurance equilibrium)**.** *If* $\max_i \beta_i(1) < F < \sum_i \beta_i(1)$*, zero contributions form a strict Nash equilibrium, although full assurance has strictly positive collective net gains. This establishes an inefficient equilibrium, not uniqueness of nonprovision.*

Proof. Starting from zero contributions by others, a deviation $0 < g_i \le F$ purchases $e = g_i/F$. Convexity and $\beta_i(0) = 0$ imply $\beta_i(e) \le e\beta_i(1) < eF = g_i$. A deviation above F is still more expensive and cannot increase the benefit above $\beta_i(1)$. Zero is therefore a strict best response for every association. The second inequality is exactly the positive collective-gain condition. □

The result does not apply when a platform can appropriate all downstream benefits, when insurance sales themselves fund the intervention through a protected franchise, or when assurance also produces sufficiently large direct productivity benefits. Those arrangements change the beneficiary and financing structure.

4.3 A conditional procurement implementation

Proposition 3 (Sequential procurement with commitment)**.** *Suppose all* $\beta_i(1)$ *are public and* $B(1) > F$*. There is a finite set of beneficiaries and fees* $f_i \in (0, \beta_i(1))$ *summing to* F *such that the following mechanism uniquely implements full assurance on the subgame-perfect equilibrium path: approach the selected beneficiaries sequentially; each accepts an enforceable fee conditional on all accepting; any rejection irrevocably cancels the project; execute only after all accept.*

Proof. Include all strictly positive beneficiaries and set $f_i = F\beta_i(1)/B(1)$. The fees sum to F and are strictly below benefits. In the final decision subgame following prior acceptances, accepting yields $\beta_i(1) - f_i > 0$, whereas rejecting yields zero. Backward induction gives strict acceptance at each earlier decision. Cancellation removes later decisions after a rejection. This proves the stated equilibrium path under commitment. □

A simultaneous unanimity offer can retain an all-reject equilibrium with at least two solicited beneficiaries: one acceptance alone does not provide the project or charge a fee. Sequential offers with credible cancellation remove that particular coordination problem. They do not establish dominant-strategy truth telling, robustness to private values, coalition proofness, or implementation if a rejected project can immediately be refinanced. The distinction matters for any institutional recommendation.

4.4 When stronger tail-risk pricing reduces assurance incentives

For this comparison, fix the same project (u_0, u_1, F) and identical willingness to pay w. Vary only insurers' tail-risk weight λ. Put $v = w - p$ and $d_k = r(u_k) - p$ for $k = 0, 1$. Assume $0 < d_1 < d_0$ and $v > 0$.

Proposition 4 (Nonmonotone assurance value)**.** *The benefit per association of full assurance is*

$$\beta(\lambda) = [v - \lambda d_1]_+ - [v - \lambda d_0]_+ = \begin{cases} \lambda(d_0 - d_1), & 0 \le \lambda \le v/d_0, \\ v - \lambda d_1, & v/d_0 < \lambda < v/d_1, \\ 0, & \lambda \ge v/d_1. \end{cases} \tag{15}$$

On $\lambda \in [0, 1]$, the maximum occurs at $\min\{v/d_0, 1\}$*. If $v/d_0 < 1$, the collective benefit peaks at $Nv(1-d_1/d_0)$. Under this interior-peak condition, a strictly beneficial project exists at some admissible λ if and only if*

$$F < Nv(1 - d_1/d_0). \tag{16}$$

When this holds, the values giving positive net collective gains are exactly

$$\frac{F}{N(d_0 - d_1)} < \lambda < \frac{v - F/N}{d_1}, \qquad 0 < \lambda \le 1. \tag{17}$$

Proof. Before the first threshold both insurance markets are active, so the gain is the difference in risk loadings. Between thresholds only the improved market is active. Above the second threshold neither is active. This yields (15), with a positive slope followed by a negative slope and then zero. Evaluating the peak and solving $N\beta(\lambda) > F$ on the first two segments gives the remaining claims, with the stated restriction $v/d_0 < 1$ for the interior peak and existence criterion. □

Here λ is a preference parameter, not a chosen prudential requirement. Interpreting the comparison as a regulatory reform would require a separate model of capital costs and default. The result also holds for known dependence whenever the two risk charges differ: it is an insurance participation mechanism amplified by ambiguous common tails, not an exclusively Knightian phenomenon. If full assurance eliminates the risk loading ($d_1 = 0$), the benefit stops increasing but does not eventually decline; residual tail risk is necessary for the downward branch.

5 Endogenous coverage and the return to collective safety

The discrete benchmark is useful for financing, but market closure is not required for the downward response. This section replaces the N identical associations with a continuum of potential clients of total exposure mass M. Every client has the same marginal claim probability p; variation in willingness to pay reflects preferences rather than adverse selection. Purchasing clients remain a large pool with the common-regime law. Thus quantity varies without changing the marginal loss law or creating a small-pool approximation claim.

Let $w(Q)$ be decreasing inverse demand for the covered layer. Let $C(Q)$ be the aggregate convex cost of making Q units of fully funded underwriting capacity available, with $C(0) = 0$ and $C' \ge 0$. The principal of prefunded assets is available at the claim date; C measures the opportunity and operating costs of supplying it.

It is not an expected-shortfall capital requirement. Supply is price taking, its cost curve is fixed during the comparison, and profits can include rents to scarce capacity. Define

$$H(Q) = \int_0^Q [w(z) - p]\, dz - C(Q), \qquad 0 \le Q \le M. \tag{18}$$

Assume H is twice continuously differentiable, $H'' < 0$, $v := H'(0) > 0$, and $H'(M) \le 0$. At per-unit tail surcharge $a \ge 0$, equilibrium coverage and total risk-adjusted market surplus are

$$Q(a) = \operatorname*{argmax}_{0 \le Q \le M} \{H(Q) - aQ\}, \qquad G(a) = \max_{0 \le Q \le M} \{H(Q) - aQ\}. \tag{19}$$

For an interior allocation, the clearing premium is $w(Q) = p + a + C'(Q)$. Consumer surplus plus risk-adjusted producer surplus equals $G(a)$. These are money-metric gains under the specified participation preferences, not a comparison of social welfare across different insurer preferences. Coverage is unique and continuous; it is positive for $a < v$ and zero for $a \ge v$.

5.1 An envelope condition and its economic meaning

Fix $0 < d_1 < d_0$, where $d_k = r(u_k) - p$, and keep H unchanged when varying $\lambda \in [0, 1]$. The intervention's gross market gain is

$$\mathcal{D}(\lambda) = G(\lambda d_1) - G(\lambda d_0). \tag{20}$$

Proposition 5 (Coverage response and assurance value)**.** *Under the preceding assumptions,*

$$\mathcal{D}(\lambda) = \int_{\lambda d_1}^{\lambda d_0} Q(a)\, da, \tag{21}$$

$$\mathcal{D}'(\lambda) = d_0 Q(\lambda d_0) - d_1 Q(\lambda d_1). \tag{22}$$

The initial derivative is $(d_0 - d_1)Q(0) > 0$. If $d_1 \ge v$, then $\mathcal{D}(1) = 0$ and the function has a positive maximum at an interior risk weight. This condition is sufficient, not necessary, and does not establish uniqueness of the maximum for general demand.

When both endpoint markets are strictly active and $\lambda > 0$, define the elasticity with respect to the tail surcharge*,*

$$\epsilon(a) = -\frac{aQ'(a)}{Q(a)}. \tag{23}$$

Then

$$\mathcal{D}'(\lambda) = \int_{d_1}^{d_0} Q(\lambda d)\{1 - \epsilon(\lambda d)\}\, dd. \tag{24}$$

In particular, elasticity strictly above one throughout the interval makes assurance value fall; elasticity strictly below one makes it rise. Equation (24), rather than either sufficient bound, gives the exact weighted criterion.

Proof. Strict concavity gives a unique maximizer. The envelope theorem gives $G'(a) = -Q(a)$, including at the choke price by continuity. Integrating and differentiating gives (21)–(22). Since $Q(0) > 0$, the initial derivative is positive. If $d_1 \ge v$, both endpoint quantities are zero at $\lambda = 1$; continuity and positivity near zero give an interior positive maximum. In the active range $Q'(a) = 1/H''(Q(a)) < 0$. Differentiating $dQ(\lambda d)$ with respect to d and integrating gives (24). □

Equation (21) is the area under equilibrium coverage between the improved and original surcharges. Increasing λ widens this surcharge interval but moves it into a region of lower coverage. The elasticity measures the net quantity response after both demand and costly supply adjust; it is not the usual demand elasticity with respect to the total premium. No tail plateau is required for Proposition 5: the result applies to any fixed pair of linear per-unit loadings. The actuarial model supplies their connection to ambiguous shared AI failures.

Endogenous coverage also preserves the investment result. Put $\ell(e) = r(u(e)) - p$, with $u(e)$ as in (11). For every fixed $Q \geq 0$, $H(Q) - \lambda\ell(e)Q$ is convex in e because $\ell(e)$ is concave. Its supremum over Q is convex. Hence $G(\lambda\ell(e)) - G(\lambda\ell(0)) - Fe$ has an endpoint optimum; full investment is uniquely optimal when $\mathcal{D}(\lambda) > F$, and zero when $\mathcal{D}(\lambda) < F$. This is a collective benchmark. It does not assert that an individual provider can collect all these gains.

5.2 A downturn while both markets are active

Suppose $w(Q) = p + v - bQ$, $M = v/b$, and $C(Q) = \kappa Q^2/2$, where $b, \kappa, v > 0$ and $p + v \leq 1$. Write $h = b + \kappa$. Direct market clearing gives

$$Q_k(\lambda) = \frac{[v - \lambda d_k]_+}{h}, \qquad G_k(\lambda) = \frac{[v - \lambda d_k]_+^2}{2h}. \tag{25}$$

Proposition 6 (An interior peak before market exit)**.** *In this linear-demand economy,*

$$\mathcal{D}(\lambda) = \frac{[v - \lambda d_1]_+^2 - [v - \lambda d_0]_+^2}{2h}. \tag{26}$$

If $\lambda^\dagger := v/(d_0 + d_1) < 1$, the unique maximizing risk weight is $\lambda^\dagger$, and

$$\mathcal{D}(\lambda^\dagger) = \frac{v^2(d_0 - d_1)}{2h(d_0 + d_1)}. \tag{27}$$

At the maximum, both markets are strictly active:

$$Q_0(\lambda^\dagger) = \frac{vd_1}{h(d_0 + d_1)} > 0, \qquad Q_1(\lambda^\dagger) = \frac{vd_0}{h(d_0 + d_1)} > 0. \tag{28}$$

The value begins falling before the original-market exit weight v/d_0. If $v/(d_0 + d_1) \geq 1$, its maximum on the admissible range is at one instead.

Proof. While both markets are active, differentiating (26) gives $(d_0 - d_1)[v - \lambda(d_0 + d_1)]/h$. Its unique zero is $\lambda^\dagger < v/d_0$. After the original market closes, the derivative is $-d_1[v - \lambda d_1]/h < 0$ until the improved market also closes, after which the value is zero. Substitution proves the peak and quantity formulas, including the truncation at $\lambda = 1$. □

Consumer surplus is $bQ_k^2/2$, and risk-adjusted producer surplus is $\kappa Q_k^2/2$. Their respective gains from the project are $(b/h)\mathcal{D}$ and $(\kappa/h)\mathcal{D}$, so the downturn appears in both groups' gross benefits. This does not supply an equilibrium financing rule; access to those benefits remains a separate problem. Increasing κ reduces total assurance value and weakly shrinks the set of risk weights with $\mathcal{D} > F$. Conditional on an interior peak and fixed d_0, d_1, v, b, it does not move the peak risk weight in this particular quadratic-cost case.

For an illustration, retain $p = 0.02$, $\alpha = 0.99$, $u_0 = 0.9$, and $u_1 = 0.2$, so $d_0 = 0.98$ and $d_1 = 0.392$. Use $v = 0.12$, $b = 0.0012$, $\kappa = 0.0006$, and a project cost $F = 0.7$. This is a different demand-and-cost economy from Section 10. The peak is at $\lambda^\dagger \simeq 0.08746$, with value 1.71429 and coverage masses 19.04762 and 47.61905. The original market does not close until $\lambda \simeq 0.12245$. At $\lambda = 0.15$ the project value is 1.0404; at $\lambda = 0.30$ it is only 0.0016. All values are constructed, not estimated.

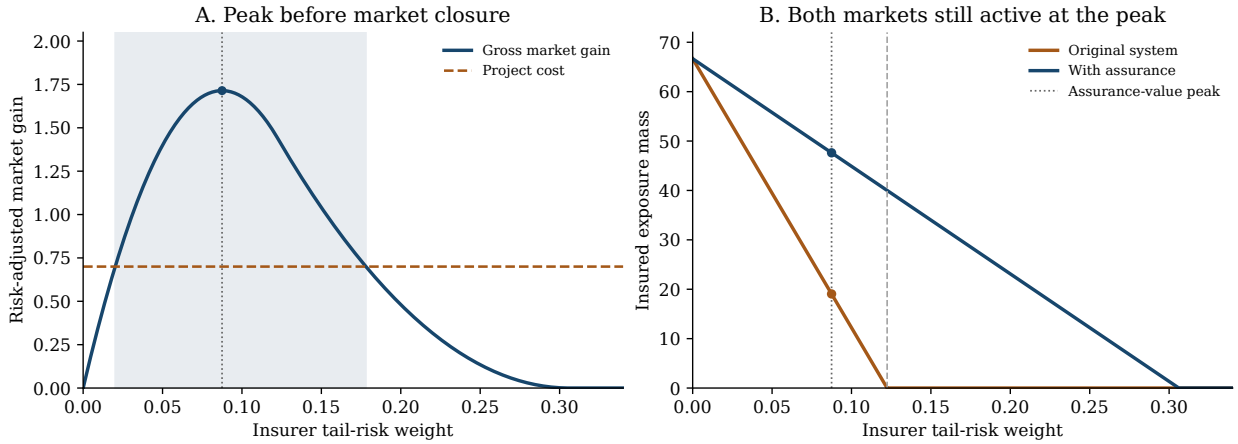


Figure 1: Continuous coverage and costly capacity. Panel A plots gross assurance value and the project cost. Panel B shows coverage before and after the intervention. The value peaks while both quantities are positive, before the original market exits. Shading marks the illustrative financing interval.

5.3 Wider ambiguity with the same physical control

Fix $\lambda \in (0, 1]$. To isolate uncertainty from an actual change in risk, fix a true common-regime probability q and a verified technology that multiplies it by $s \in (0, 1)$. Change only the underwriting bound $u \in [q, 1]$. The credible dependence intervals are $[0, u]$ without the control and $[0, su]$ with it. The actual laws, Y_q and Y_{sq}, do not change as u changes. Define

$$\mathcal{A}(u) = G\big(\lambda[r(su) - p]\big) - G\big(\lambda[r(u) - p]\big). \tag{29}$$

Write $A = p(1 - p)/\tau$. Before saturation, $u < u_*$, the derivative is

$$\mathcal{A}'(u) = \lambda A\{Q(\lambda Au) - sQ(\lambda Asu)\}. \tag{30}$$

The same weighted-coverage comparison therefore applies. In the linear-demand economy, if

$$q < u^\dagger := \frac{v}{\lambda A(1 + s)} < u_*, \tag{31}$$

then $\mathcal{A}$ has its unique global maximum on $[q, 1]$ at $u^\dagger$. Both markets remain active there.

To verify the claim, substitute the two linear loadings Au and Asu into (26). Up to u_* the derivative changes sign at $u^\dagger$, before either market exits. Once $u \geq u_*$, the original loading is constant while the improved loading is nondecreasing, so $\mathcal{A}$ cannot increase. The strict decrease immediately after $u^\dagger$ establishes uniqueness. Thus admitting more pessimistic dependence models can lower the financeable benefit of the same control, even though each premium weakly rises. It is not an argument for discarding credible adverse models.

For the preceding parameters, fix $\lambda = 0.15$, $s = 2/9$, and $q = 0.2$. The physical dependence falls from 0.2 to 0.04444 under the control in every comparison. Market value rises from approximately 2.13592 at $u = q$ to 2.54545 at $u^\dagger \simeq 0.33395$, then falls to 1.0404 at $u = 0.9$. This comparison makes ambiguity, rather than a changing catastrophe frequency, the source of the shift in market incentives.

5.4 Tail weight and tail focus are different comparative statics

The preceding results vary the weight λ on a fixed tail statistic. Increasing the ES confidence level α instead changes which losses the statistic distinguishes. This produces a second mechanism that does not require insurance-market exit. A control can make a catastrophic payout less likely without changing its amount. Sufficiently extreme tail pricing can then assign the original and improved systems the same charge.

The following result does not require the common-regime family. Let $\mathcal{P}_0, \mathcal{P}_1$ be nonempty sets of laws for a normalized insured loss $X \in [0, 1]$. Every law has mean p, and no prior is assigned to either set. Define

$$R_k(\alpha) = \sup_{P \in \mathcal{P}_k} \mathrm{ES}_\alpha^P(X), \qquad b_k = \sup_{P \in \mathcal{P}_k} \operatorname{ess\,sup}_P X.$$

The control is assumed to lower robust ES at every confidence level: $R_1(\alpha) \le R_0(\alpha)$. This is a restriction on its effect, not a consequence of holding marginal losses fixed. Keep demand, capacity costs, and $\lambda > 0$ fixed. Its market value is

$$V(\alpha) = G\big(\lambda[R_1(\alpha) - p]\big) - G\big(\lambda[R_0(\alpha) - p]\big).$$

Proposition 7 (A common worst payout and vanishing assurance value)**.** *The function V is nonnegative and continuous on $[0, 1)$, with $V(0) = 0$, and*

$$\lim_{\alpha \uparrow 1} V(\alpha) = G\big(\lambda[b_1 - p]\big) - G\big(\lambda[b_0 - p]\big). \tag{32}$$

If $b_0 = b_1 = b$ and V is positive at some confidence level, its positive global maximum is attained at an interior confidence level and its value tends to zero at the extreme tail. Uniqueness or a single-peaked shape is not asserted. If also $\lambda(b - p) < v = H'(0)$, both counterfactual insurance markets remain strictly active at every confidence level, including in the limit.

Proof. For bounded loss, the quantile representation gives $\mathrm{ES}_0^P(X) = p$ and $\mathrm{ES}_\alpha^P(X) \uparrow \operatorname{ess\,sup}_P X$. The supremum over laws commutes with the increasing limit: an approximately maximizing fixed law supplies the lower bound, and b_k supplies the upper bound. ES is uniformly continuous in α over each compact subinterval of $[0, 1)$, uniformly over laws supported on $[0, 1]$. This follows from its infimum representation with the minimizing threshold restricted to $[0, 1]$. Taking a supremum therefore preserves continuity. Composing with continuous G proves the limit. If the endpoints coincide, extending $V(1) = 0$ and applying compactness gives an interior positive maximizer. Finally, all surcharges are at most $\lambda(b - p) < v$, so all equilibrium quantities are positive. □

The ES endpoint and quantile properties are standard (Acerbi and Tasche, 2002, Section 3); weighting quantiles to represent tail concern is also established (Acerbi, 2002, Sections 2–3). The economic application separates two ways assurance value can fall. With greater λ, coverage contracts relative to the saving per exposure. With greater α, the pricing statistic itself can cease to distinguish a frequency reduction. The latter result applies even to fixed positive coverage, for which the gross saving is $\lambda Q[R_0(\alpha) - R_1(\alpha)]$. A control that instead reduces the largest admitted payout need not lose its extreme-tail value. Neither statement evaluates uninsured social harm or recommends choosing a less conservative criterion.

5.5 The residual dependence bound required to cover project cost

Return to the common-regime model and write $\tau = 1 - \alpha$. Fix $u_0 > 0$ and consider a verified control delivering an upper bound $u_1 \in [0, u_0]$. When $\tau < u_0 p$, the original risk charge is one and the improved loading is

$$d_1(\tau, u_1) = \min\left\{1 - p, \frac{u_1 p(1-p)}{\tau}\right\}.$$

For every fixed $u_1 > 0$, both charges equal one once $\tau \le u_1 p$. A large proportional reduction in common risk can therefore generate zero insurance-price saving at a sufficiently extreme confidence level.

There is also an exact funding test. Suppose $0 < \lambda(1-p) < v$, so both markets stay active, and let

$$0 < F < G(0) - G\big(\lambda(1-p)\big).$$

Define the unique $a_F \in (0, \lambda(1-p))$ by

$$G(a_F) = F + G\big(\lambda(1-p)\big).$$

Strict monotonicity of G on this interval gives

$$V(1-\tau) > F \quad \Longleftrightarrow \quad u_1 < \frac{a_F}{\lambda p(1-p)}\,\tau, \qquad \tau < u_0 p. \tag{33}$$

Equality gives zero net market gain. The admissible residual bound must thus shrink at least on the scale of the underwriting tail probability, with a stricter coefficient for a more expensive project. This compares candidate bounds at a fixed project cost; it does not specify the cost of attaining each bound. It is a test of whether total market gains cover cost, not a mechanism for collecting those gains. Nor does it imply that an audit can always deliver the required bound: a true residual common-regime probability above it cannot be removed by more favorable reporting.

Appendix I illustrates a fixed control, gives the joint limit when the residual bound shrinks with τ, and examines finite pools. Eliminating the common regime leaves an independent finite pool with a positive probability that all clients claim. Its extreme-tail limit differs from the result obtained by taking the large-pool limit first. The scope of a large-pool approximation must therefore specify confidence level as well as pool size. The next subsection makes the cost of obtaining a tighter bound endogenous.

5.6 Costly control: greater effort followed by abandonment

The fixed-cost frontier leaves the cost of achieving a tighter bound unspecified. We now let the collective decision maker choose physical effort. This replaces the linear technology in Section 4 for this subsection. Write $A = \lambda(1-p)$ and assume $0 < A < v$, so insurance remains active under every feasible control. Let $u(e) \in (0, u_0]$ be continuous and nonincreasing for $e \ge 0$, with $u(0) = u_0 > 0$. A verified intervention reduces every admitted common-regime probability proportionally to $u(e)/u_0$, preserving each client's marginal loss probability. Its resource cost $K(e)$ is continuous, satisfies $K(0) = 0 < K(e)$ for $e > 0$, and tends to infinity with effort. Some common risk therefore remains at every finite effort.

For $\tau \le u_0 p$, the original surcharge is A. The collective decision maximizes

$$W_\tau(e) = G\left(\lambda \min\left\{1 - p, \frac{u(e)p(1-p)}{\tau}\right\}\right) - G(A) - K(e). \tag{34}$$

This is aggregate market gain net of resources, before any constraint on collecting the gain. Define the maximum possible gross gain $B_{\max} = G(0) - G(A) > 0$.

Proposition 8 (Eventual abandonment with positive residual common risk)**.** *Let $u_B = \min\{u(e) : K(e) \leq B_{\max}\}$. Then $u_B > 0$, and zero effort is uniquely optimal whenever $0 < \tau \leq pu_B$. Both the original and improved insurance markets remain active. In contrast, if a feasible finite-cost intervention eliminates the common regime and costs less than $B_{\max}$, zero effort is not optimal for any $\tau < u_0 p$.*

Proof. The cost sublevel set is nonempty and compact. Continuity and strictly positive residual risk give $u_B > 0$. On this set, $\tau \leq pu_B$ makes the maximum-payout atom fill the assessed tail under every affordable intervention, so gross gain is zero. Every positive effort then has negative net gain. Outside the set, cost exceeds the maximum possible gross gain. Complete common-regime removal instead gives surcharge zero in the large-pool model and net gain $B_{\max} - K > 0$. □

An exponential technology gives the entire global choice in the saturated-baseline region:

$$u(e) = u_0 \exp(-\beta e), \qquad K(e) = ce, \qquad \beta, c > 0.$$

It can be implemented by replacing the common regime with independent draws with probability $1 - \exp(-\beta e)$. The transformation holds for every admitted initial parameter; it is not an assumption of reassuring audit news. Put $\chi = c/\beta$ and $z = \log(u_0 p/\tau) > 0$. Effort up to z/β leaves the premium unchanged. Define

$$J(\chi) = \max_{y \geq 0} \{G(Ae^{-y}) - G(A) - \chi y\}, \qquad \mathcal{Y}(\chi) = \operatorname*{argmax}_{y \geq 0} \{G(Ae^{-y}) - G(A) - \chi y\}. \tag{35}$$

The maximum exists and $J \geq 0$. Neither object depends on τ or u_0.

Proposition 9 (Intensification and the global abandonment threshold)**.** *Under the exponential technology,*

$$\max_{e \geq 0} W_\tau(e) = \max\{0, J(\chi) - \chi \log(u_0 p/\tau)\}. \tag{36}$$

If $J = 0$, zero effort is uniquely optimal for $\tau < u_0 p$. If $J > 0$, define

$$\tau_* = u_0 p \exp[-J(\chi)/\chi]. \tag{37}$$

For $\tau_* < \tau < u_0 p$, the optimal efforts are exactly $e^* = [\log(u_0 p/\tau) + y^*]/\beta$, with $y^* \in \mathcal{Y}(\chi)$. For $\tau < \tau_*$, zero effort is uniquely optimal. At $\tau = \tau_*$, zero and these strictly positive efforts are optimal.

Proof. For $\beta e \leq z$, net gain is $-ce$. For $\beta e \geq z$, set $y = \beta e - z$. The surcharge becomes Ae^{-y}, so net gain is $G(Ae^{-y}) - G(A) - \chi y - \chi z$. Maximizing this expression and comparing it with zero proves the statements. Bounded gross gain and $\chi > 0$ ensure that the inner objective tends to minus infinity. If $J > 0$, every inner maximizer is strictly positive. □

Conditional on a fixed positive branch, effort rises by $1/\beta$ for a unit increase in $\log(1/\tau)$. The residual bound is $u(e^*) = (\tau/p)e^{-y^*}$; the improved surcharge Ae^{-y^*} and coverage are constant on that branch. More effort is needed to earn the same gross market gain. Eventually its cost absorbs that gain, and the optimal choice jumps to zero. This is a comparative static across risk criteria, not a prediction of behavior over calendar time. General demand may yield several positive branches; the proposition characterizes their full set.

Wider dependence ambiguity has a parallel conditional effect. At fixed true initial dependence, p, τ, and physical technology, raising the admitted upper bound u_0 increases required effort by $1/\beta$ per unit of $\log u_0$ while a positive branch remains optimal. It reduces optimized positive net gain at rate χ and can induce abandonment. This varies the ambiguity set, not ambiguity aversion or the true risk process. The comparison remains within $\tau < u_0 p$ and $u_0 \leq 1$.

Figure 2 illustrates the global decision. Appendix I solves the linear-demand case, relates its curvature to coverage elasticity, and checks the no-investment option explicitly. The exact zero-investment result uses saturation at a maximum-payout atom. The bounded-law limit in Proposition 7, by itself, does not justify uniform convergence over a changing set of controls or exact abandonment. These results concern the insurance channel; uninsured safety benefits and decentralized payment incentives can change the decision.

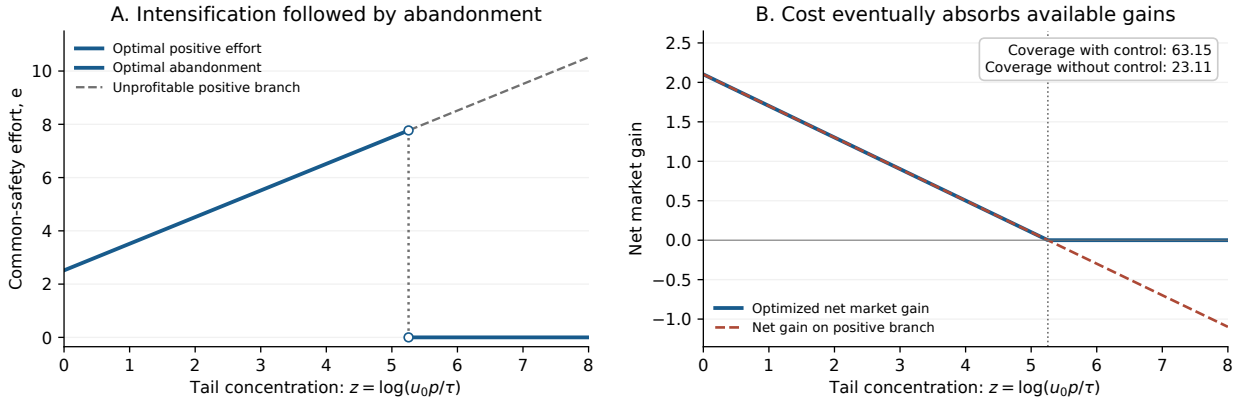


Figure 2: Costly common-safety control under more extreme tail focus. The positive branch requires increasing effort to preserve its gross market gain. The optimum switches to zero when cost absorbs that gain; the two marked efforts tie at the threshold. Coverage remains strictly positive with or without control. The horizontal axis varies the assessed tail probability, not time. All inputs are constructed.

5.7 The insurance channel does not exhaust safety value

Suppose total deployment mass M stays fixed when coverage changes. At the common-regime aggregation limit, clients retain liability $(M - Q_k)Y$ under regime k, whose worst-case expected shortfall is

$$R_k = (M - Q_k)r(u_k). \tag{38}$$

If both insurance markets close, $\mathcal{D} = 0$, yet $R_0 - R_1 = M[r(u_0) - r(u_1)] > 0$. The verified control still improves the bound on retained aggregate tail loss. This is a risk-exposure statistic, not an additional welfare term to add to $\mathcal{D}$, and not a claim about the effect under every true dependence parameter. The theorem concerns returns available through the specified insurance market. It cannot support a conclusion that stronger prudential regulation reduces social safety or that a project without insurance-market funding has no social benefit.

6 Private benefits and the financing constraint

The previous section varies aggregate coverage under a known demand curve. We now return to the constant-unit-price association benchmark to isolate a different problem: the beneficiaries know their own valuations, but the coordinator does not. These are separate extensions. A finite number of associations still contains a large population of underlying insured exposures. Insurers earn zero risk-adjusted rents in this benchmark; their expected risk compensation is not an unassigned pool of funds for assurance.

6.1 How insurance participation changes the information problem

There are $N \geq 2$ associations, with privately known net willingness to pay $v_i = w_i - p$ in the public interval $[\underline{v}, \overline{v}]$, where $0 < \underline{v} < \overline{v} \leq 1 - p$. No prior over these types is assumed. There is no verifiable purchase history. We first procure assurance before separate competitive insurance transactions, using project fees conditional on reports and provision. Section 6.4 then allows the mechanism to allocate insurance and collect all payments jointly. The project changes the publicly assessed loadings from d_0 to d_1, with $0 < d_1 < d_0$, at cost $F > 0$. Association i's incremental benefit is

$$z_i = b_\lambda(v_i) := [v_i - \lambda d_1]_+ - [v_i - \lambda d_0]_+. \tag{39}$$

The attainable benefit domain is the entire interval $[\ell_\lambda, h_\lambda]$, where

$$\ell_\lambda = b_\lambda(\underline{v}), \qquad h_\lambda = b_\lambda(\overline{v}). \tag{40}$$

Continuity and monotonicity of b_λ give this domain even though the mapping can have flat segments. Types with the same z_i have identical preferences over the project and monetary fees.

For $\lambda d_0 \leq \underline{v}$, every possible type buys insurance before and after the project. All receive the same saving:

$$\ell_\lambda = h_\lambda = \lambda(d_0 - d_1). \tag{41}$$

Private willingness to pay is then irrelevant to project financing. When $\lambda d_0 > \underline{v}$, some types no longer buy the original insurance. On the range

$$\frac{\underline{v}}{d_0} < \lambda < \min\left\{\frac{\underline{v}}{d_1}, \frac{\overline{v}}{d_0}\right\}, \tag{42}$$

the benefit endpoints are $\ell_\lambda = \underline{v} - \lambda d_1$ and $h_\lambda = \lambda(d_0 - d_1)$. Greater tail sensitivity lowers the guaranteed benefit while increasing the largest benefit. It makes existing private information economically relevant; it does not create new information about the loss process.

6.2 A sharp funding benchmark

For a fixed λ, abbreviate the benefit endpoints as ℓ and h. A direct mechanism chooses provision $x(\mathbf{z}) \in \{0, 1\}$ and payments $t_i(\mathbf{z})$ to the coordinator. Incremental utility is $z_i x - t_i$. We impose three requirements:

1. Efficient project selection at every profile: $x^*(\mathbf{z}) = 1$ if and only if $\sum_i z_i \geq F$, with provision at equality.
2. Dominant-strategy incentive compatibility: truthful reporting maximizes utility for every report profile of the other associations.

3. Ex post individual rationality: $z_i x^*(\mathbf{z}) - t_i(\mathbf{z}) \geq 0$ at every true profile.

The outside option is the existing insurance opportunity with no assurance project. This participation condition is weaker than withdrawal stability when an outsider can enjoy a project financed by others. We do not assume that truthful reporting is the only equilibrium.

Let S^* be the smallest uniform bound on outside funding across mechanisms satisfying these requirements:

$$S^* = \inf_{\{t_i\}} \sup_{\mathbf{z}\in[\ell,h]^N} \left[Fx^*(\mathbf{z}) - \sum_i t_i(\mathbf{z}) \right]_+ . \tag{43}$$

This is a worst-profile financing requirement, not an expected subsidy. It requires neither a prior over private benefits nor a probability assigned to the worst profile.

Proposition 10 (Critical payments and the funding bound)**.** *For $N \geq 2$,*

$$S^* = \begin{cases} 0, & F \leq N\ell \text{ or } F \geq Nh, \\ F - N\max\{\ell, F-(N-1)h\}, & N\ell < F < Nh. \end{cases} \tag{44}$$

In the intermediate regime the bound is strictly positive and can also be written as

$$S^* = \min\{F - N\ell,\ (N-1)(Nh-F)\}. \tag{45}$$

Consequently, efficient selection at every profile, dominant-strategy truth telling, individual rationality, and zero outside funding cannot all hold in that regime.

Proof. Fix the other reports and write $c_i = F - \sum_{j\neq i} z_j$. If $c_i \leq \ell$, the efficient rule provides at every own report. Incentive compatibility makes the payment constant in that report, and individual rationality at ℓ bounds it by ℓ. If $\ell < c_i \leq h$, the efficient rule switches at c_i. Incentive compatibility on the continuous type interval makes the difference between the provision and nonprovision payments equal to c_i. Individual rationality bounds a nonprovision payment by zero. Hence, whenever provision occurs,

$$t_i(\mathbf{z}) \leq \max\{\ell, F - \textstyle\sum_{j\neq i} z_j\}. \tag{46}$$

At $\mathbf{z} = (h, \ldots, h)$ this gives the lower bound in the intermediate case of (44). Both ℓ and $F-(N-1)h$ are strictly below F/N there, proving positivity.

The bound is attained by setting the payment equal to the right side of (46) when the project is provided and zero otherwise. These critical payments are incentive compatible and individually rational. Among provision profiles each payment is minimized when all other benefits are h, so the all-high profile generates the largest deficit. For $F \leq N\ell$, constant provision and a fee F/N for each association satisfy all requirements without outside funds. For $F > Nh$ the project is never provided; at $F = Nh$ it can be funded at the all-high profile by fees h. Finally, subtracting the maximum in (44) gives (45). □

The incentive argument is standard public-project logic, specialized to the insurance-generated benefit domain. The economic object of interest is its dependence on λ, ℓ_λ, and h_λ. Compulsory transfers, verifiable type information, inefficient allocations, or a Bayesian incentive criterion change the problem. Bundling insurance with assurance requires a separate check, provided below. No universal impossibility of funding AI safety follows.

6.3 More collective gain, a larger financing requirement

Proposition 11 (Tail pricing can raise gains and the funding requirement together)**.** *Consider an open interval of admissible risk weights satisfying* (42) *and*

$$N\ell_\lambda < F < \ell_\lambda + (N-1)h_\lambda. \tag{47}$$

On that interval,

$$\frac{dS^*(\lambda)}{d\lambda} = Nd_1 > 0, \tag{48}$$

$$\frac{d}{d\lambda}\{Nh_\lambda - F\} = N(d_0 - d_1) > 0. \tag{49}$$

Thus the minimum uniform outside funding rises while net project gains rise at the fixed willingness-to-pay profile $v_i = \overline{v}$ for every i.

Proof. On (42), $\ell_\lambda = \underline{v} - \lambda d_1$ and $h_\lambda = \lambda(d_0 - d_1)$, with $\ell_\lambda < h_\lambda$. Condition (47) implies $N\ell_\lambda < F < Nh_\lambda$ and $F - (N-1)h_\lambda < \ell_\lambda$. Therefore (44) becomes $S^*(\lambda) = F - N\underline{v} + N\lambda d_1$. Differentiating this expression and $Nh_\lambda - F$ proves the claim. □

The decline in the lowest benefit limits what can be collected from a high-benefit association when its peers' reports make it nonpivotal. More favorable realizations of aggregate benefits do not remove the incentive restrictions imposed by other possible type profiles. This mechanism differs from a decline in project surplus: both derivatives in Proposition 11 are positive.

For an illustration, use $d_0 = 0.98$, $d_1 = 0.392$, $N = 3$, $\underline{v} = 0.06$, $\overline{v} = 0.16$, and $F = 0.06$. The risk charges correspond to the same actuarial parameters as before; the number of associations, willingness-to-pay range, and project cost define a separate example. The following quantities are constructed.

λ	ℓ_λ	h_λ	$[Nh_\lambda - F]_+$	$S^*(\lambda)$
0.04	0.02352	0.02352	0.01056	0.00000
0.12	0.01296	0.07056	0.15168	0.02112
0.14	0.00512	0.08232	0.18696	0.04464
0.20	0.00000	0.08160	0.18480	0.06000

Increasing λ from 0.12 to 0.14 raises net gains at the all-high profile from 0.15168 to 0.18696, but more than doubles the smallest uniform funding bound. High types remain insured both before and after assurance at both risk weights. At $\lambda = 0.20$, the efficient truthful mechanism can require the full project cost from outside the beneficiaries.

This does not mean that the all-high project cannot be privately funded. A conditional unanimity rule charging F/N to each association provides whenever every reported benefit is at least F/N. It is truthful in dominant strategies and individually rational, with no outside funding. Its cost is inefficient nonprovision at some mixed profiles. At $\lambda = 0.12$, for example, the profile $(\ell_\lambda, h_\lambda, h_\lambda)$ has total benefit $0.15408 > F$, but the low-benefit association rejects its 0.02 share. The funding bound concerns efficiency at *every* profile, not the cost of financing a single known profile.

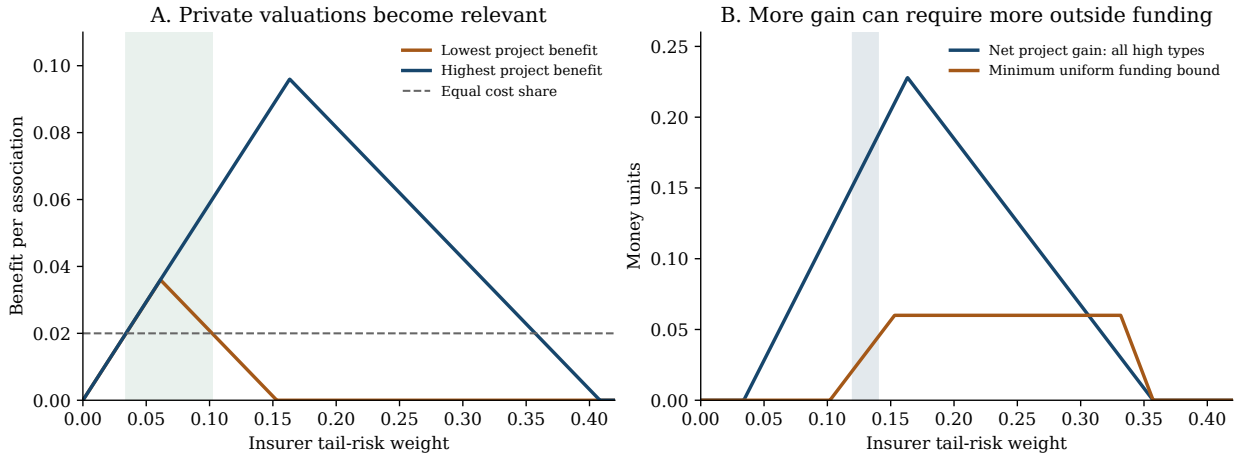


Figure 3: Private benefits and the financing constraint. Panel A shows the benefit domain and the equal cost share; shading marks the interval in which even the lowest benefit covers that share. Panel B distinguishes net project gains at the all-high profile from the minimum uniform outside funding needed for an efficient, truthful, individually rational mechanism. The highlighted comparison is $\lambda = 0.12$ to 0.14. All parameters are illustrative.

Outside funding must come from an identified source. It cannot be supplied by relabeling a tail-risk charge as available capital or counting reinsurance receivables as new assets. An insurer or platform with additional appropriable gains could be included as another beneficiary, changing the benefit domain. Likewise, a mandatory levy changes the participation constraint. These institutional choices are alternatives to the stated benchmark, not free consequences of stronger risk pricing.

6.4 Bundled insurance, efficient coverage, and observed purchases

Could a coordinator remove the financing constraint by selling insurance and assurance together? We now allow a direct contract to choose the project $x(\mathbf{v})$, allocate the full insured layer $q_i(\mathbf{v}) \in \{0, 1\}$, and collect a single payment from each association. Let $a_k = \lambda d_k$, and let P_i denote the payment after subtracting the expected-claim component pq_i. Utility relative to remaining uninsured is $v_i q_i - P_i$. The no-project outside option remains

$$O(v_i) = [v_i - a_0]_+. \tag{50}$$

These are full-layer allocations; the argument does not assume linear utility for arbitrary deductibles or partial indemnities.

Efficient coverage conditional on the project is $q_i^*(\mathbf{v}) = \mathbf{1}\{v_i \geq a_{x^*(\mathbf{v})}\}$, with coverage at indifference. Maximizing the specified risk-adjusted surplus over both allocations gives the same project rule as before: $x^* = 1$ if and only if $\sum_i b_\lambda(v_i) \geq F$. Bundled contracts must satisfy dominant-strategy incentive compatibility in v_i and participation $v_i q_i^* - P_i \geq O(v_i)$. Their uniform funding requirement is minimized over all such payments:

$$S_{\mathrm{B}}^* = \inf_{\{P_i\}} \sup_{\mathbf{v} \in [\underline{v}, \overline{v}]^N} \left[F x^* + \sum_i a_{x^*} q_i^* - \sum_i P_i \right]_+ . \tag{51}$$

The $a_{x^*}q_i^*$ terms reimburse insurers' specified monetary risk compensation. Expected claims have already been subtracted from both receipts and required payments. Neither term is identified with regulatory capital.

Proposition 12 (Bundling does not remove the efficient-allocation funding bound)**.** *Under the same type domain and no-project participation benchmark,*

$$S_{\mathrm{B}}^* = S^*. \tag{52}$$

Thus Proposition 11 continues to hold when a single contract allocates insurance and assurance, provided coverage and project selection are efficient at every type profile.

Proof. First, the separate critical-payment mechanism is feasible in the bundled class. Set $P_i = a_{x^*}q_i^* + t_i$, using the payments from Proposition 10. Truthful coverage maximizes $v_i q - a_{x^*}q$. Under a false report the imposed coverage can only lower this value relative to choosing coverage freely. The original project mechanism already prevents profitable deviations with free coverage choice. Hence the bundle is incentive compatible, satisfies participation, and has the same funding bound.

For the reverse inequality, only $N\ell < F < Nh$ needs examination. Fix all other types at $\overline{v}$, and put $c = F - (N-1)h < h$. At the all-high profile the project and every insurance layer are provided. If $c \le \ell$, the project is provided at every own report. When $\underline{v} \ge a_1$, coverage is always one, so incentive compatibility makes P_i constant across own reports. Participation at $\underline{v}$ gives

$$P_i - a_1 \le \underline{v} - a_1 - O(\underline{v}) = \ell.$$

When $\underline{v} < a_1$, coverage instead switches at a_1. Incentive compatibility on the continuous type interval fixes the payment jump at a_1, while participation bounds the uncovered type's payment by zero. Thus $P_i - a_1 \le 0 = \ell$.

If $\ell < c < h$, the project switches at $v_c = a_1 + c$, which lies strictly between $\underline{v}$ and $\overline{v}$ and below a_0. Efficient coverage is zero below v_c and one above it. The payment jump must therefore equal v_c. Participation for an uncovered type bounds its payment by zero, giving $P_i - a_1 \le c$ at the all-high profile. In either case net project receipts from each association are at most $\max\{\ell, c\}$. Substitution into (51) yields the lower bound S^* from (44). In the remaining cost regimes the feasible separate mechanism needs no outside funding, which proves equality there as well. □

This result allows arbitrary monetary transfers within the specified joint allocation problem. It does not cover screening through a different coverage menu, costly preliminary consumption, or contracts with a different participation baseline. A staged payment or purchase commitment that merely implements the same truthful full-layer allocation cannot evade its payment restrictions. If avoiding the funding bound requires withholding efficient insurance, rejecting an efficient project, or imposing an additional participation cost, that change must be included in the economic comparison.

Verifiable earlier purchases can supply genuinely new information. Suppose an association bought the same full layer for a previous, comparable coverage period at price $p + a_0$, before the financing mechanism was anticipated, and its valuation remains unchanged. Under the maintained choice model this establishes $v_i \ge a_0$, so its benefit for the new coverage period is the known saving $a_0 - a_1$. If this is established for all associations, a beneficial project is funded by fixed fees $F/N \le a_0 - a_1$; the private-benefit interval has collapsed. This comparison uses a new insurance purchase, not a refund of a sunk premium. The observation must be informative under the actual timing and incentives. A purchase induced by the proposed mechanism is an endogenous signal, not costless evidence of the same inequality.

7 Endogenous architecture and the diversification discount

The preceding models hold the source of shared exposure fixed except for a common control. This section permits firms to choose that source. It is a separate benchmark: all firms buy insurance, the two provider failure processes are fixed, and adoption determines the aggregate exposure to each. This separates an architecture response from the contraction in coverage studied in Section 5.

7.1 A joint book with ambiguous provider dependence

A unit mass of firms chooses between two observable models. Every user of model j loses the insured unit when $Z_j = 1$. Each Z_j is Bernoulli with the same known probability p, where

$$0 < p < \tfrac{1}{2}, \qquad p^2 < \tau < p. \tag{53}$$

The joint-failure probability is $t = \Pr(Z_1 = Z_2 = 1)$. Insurers admit $t \in [p^2, \bar{t}]$, with $p^2 \le \bar{t} \le p$, without a prior on t. The four state probabilities are t, $p - t$, $p - t$, and $1 - 2p + t$. Unlike the common-regime parameter in Section 3, t is a joint probability; its associated correlation is $(t - p^2)/[p(1 - p)]$. This is a different dependence family, not a relabeling of θ.

Firm b receives a deterministic productivity gain $b \ge 0$ from model 1 relative to model 2. The distribution H is continuous on $[0, \overline{b}]$, has a continuous positive density on its interior, and has median $m = H^{-1}(1/2)$. Each firm has the same full-layer willingness to pay w, independent of b and of the provider choice. CARA preferences and identical marginal losses support this restriction. Model fees are included in b, and firms are price takers. Restrict the comparative statics to risk weights satisfying $w > p + \lambda(1 - p)$, so insurance participation is complete at all equilibrium quotes below.

For an insured book $\mathbf{x} = (x_1, x_2) \ge 0$, liability is $x_1 Z_1 + x_2 Z_2$. The insurer's reservation compensation is

$$C(\mathbf{x}) = (1 - \lambda)p(x_1 + x_2) + \lambda \sup_{p^2 \le t \le \bar{t}} \mathrm{ES}^t_\alpha(x_1 Z_1 + x_2 Z_2). \tag{54}$$

All maximum claims remain prefunded. The notation C denotes monetary reservation compensation, not an expected-shortfall capital requirement.

Lemma 2 (Provider concentration and tail saturation)**.** *Let* $k = \min\{\bar{t}/\tau, 1\}$. *For every nonnegative book,*

$$\sup_t \mathrm{ES}^t_\alpha(x_1 Z_1 + x_2 Z_2) = \max\{x_1, x_2\} + k \min\{x_1, x_2\}, \tag{55}$$

$$C(\mathbf{x}) = c_H \max\{x_1, x_2\} + c_L \min\{x_1, x_2\}, \tag{56}$$

$$c_H = (1 - \lambda)p + \lambda, \qquad c_L = (1 - \lambda)p + \lambda k.$$

Proof. Suppose $x_1 \ge x_2$. If $t \ge \tau$, the joint-loss state fills the tail, yielding $x_1 + x_2$. If $t < \tau$, that state contributes $t(x_1 + x_2)$ and the remaining tail is filled by the model-1-only state with loss x_1. There is enough probability for this because $p > \tau$. The tail average is $x_1 + (t/\tau)x_2$, increasing in t. Evaluating at $\bar{t}$ and using symmetry proves both equations, including zero and equal exposures. □

Competitive carriers share this assessment and may pool both providers. Per-unit quotes $\boldsymbol{\pi}$ support C: they satisfy $\boldsymbol{\pi} \cdot \mathbf{y} \leq C(\mathbf{y})$ for all nonnegative books $\mathbf{y}$, with equality at the supplied book. Thus no alternative portfolio earns positive risk-adjusted profit. When $x_1 > x_2 > 0$, the quotes are (c_H, c_L). Their difference

$$\delta = c_H - c_L = \lambda(1 - k) \tag{57}$$

compensates for the diversification supplied by the smaller exposure. At equal shares the admissible quote differences range from $-\delta$ to δ, with quote sum $c_H + c_L$. Pricing each provider's book in isolation would eliminate this credit by assumption; here carriers may hold both books.

7.2 Opposite effects of tail weight and dependence ambiguity

Proposition 13 (Architecture choice under competitive insurance)**.** *The equilibrium share using model 1 is*

$$s^*(\lambda, \bar{t}) = \max\left\{\tfrac{1}{2},\ 1 - H\big(\lambda[1 - \min\{\bar{t}/\tau, 1\}]\big)\right\}, \tag{58}$$

where H is extended by zero and one outside its support. If $\delta < m$, supporting quotes are (c_H, c_L). If $\delta \geq m$, the share is $1/2$ and supporting quotes are $(c_H + c_L + m)/2$ and $(c_H + c_L - m)/2$.

For $\bar{t} < \tau$ and an interior share $1/2 < s^* < 1$,

$$\frac{\partial s^*}{\partial \bar{t}} = \frac{\lambda}{\tau} H'(\delta) > 0, \qquad \frac{\partial s^*}{\partial \lambda} = -(1 - \bar{t}/\tau) H'(\delta) < 0. \tag{59}$$

Globally, the share is weakly increasing in $\bar{t}$ and weakly decreasing in λ. If $\bar{t} \geq \tau$, all firms use model 1.

Proof. All firms insure by the maintained willingness-to-pay restriction. A firm chooses model 1 exactly when $b \geq \pi_1 - \pi_2$. A share below $1/2$ cannot clear: model 1 would then carry the lower quote, and every positive-b firm would prefer it. Above $1/2$, supporting quotes differ by δ, giving $s = 1 - H(\delta)$. This exceeds $1/2$ exactly when $\delta < m$. Otherwise equal shares require the median quote difference m, which belongs to the supporting interval because $m \leq \delta$. The stated balanced quotes meet both this difference and the zero-profit sum. These observations prove existence and the unique aggregate share. Differentiation gives the interior comparative statics. At $\bar{t} \geq \tau$, $\delta = 0$ and $H(0) = 0$. □

More severe tail pricing and wider dependence ambiguity therefore need not have the same effect on architecture. The first increases the relative reward for a diversifying exposure; the second makes the reward less credible. Saturation requires only $\bar{t} \geq \tau$, corresponding to a correlation bound $(\tau - p^2)/[p(1-p)] < 1$. Perfect correlation need not be admitted for the diversification incentive to disappear.

This is a competitive allocation, not a demonstrated externality. For a given share s, assign model 1 to the highest-b firms and write their total productivity gain as $J(s) = \int_{H^{-1}(1-s)}^{\overline{b}} b\, dH(b)$. The allocation maximizes $J(s) - C(s, 1-s)$. Indeed, $J'(s) = H^{-1}(1-s)$; the compensation derivative is δ above $1/2$ and $-\delta$ below it, yielding exactly (58). The optimum uses the maintained risk-adjusted criterion and ignores no priced portfolio effect. A claim of excessive concentration would require a different welfare benchmark, unpriced harms, or another friction.

7.3 The true tail can worsen without a change in model reliability

Hold the actual provider law fixed at $t_0 \in [p^2, \tau)$ and vary only the underwriting bound over $\bar{t} \geq t_0$. At an equilibrium share $s^* \geq 1/2$, aggregate covered loss has actual expected shortfall

$$R_{\text{true}}(\bar{t}) = s^* + \frac{t_0}{\tau}(1 - s^*). \tag{60}$$

It is strictly increasing in the ambiguity bound whenever the share is interior:

$$\frac{dR_{\text{true}}}{d\bar{t}} = \left(1 - \frac{t_0}{\tau}\right)\frac{\lambda}{\tau}H'(\delta) > 0. \tag{61}$$

Individual loss probability and aggregate expected loss remain p. The changed tail is generated by firms' exposure choices, not by a deterioration in either model's reliability or by insurance exit. The statement concerns an actual-law risk statistic, not an estimate of social welfare or an argument for ignoring credible adverse dependence models.

For a constructed example, let $p = 0.02$, $\tau = 0.01$, $\lambda = 0.1$, $w = 0.2$, b be uniform on $[0, 0.12]$, and $t_0 = p^2 = 0.0004$. All firms insure because $c_H = 0.118 < w$.

$\bar{t}$	Model-1 share	π_1	π_2	Actual aggregate ES
0.002	0.5000	0.108	0.048	0.52
0.006	0.6667	0.118	0.078	0.68
0.009	0.9167	0.118	0.108	0.92
0.010	1.0000	0.118	0.118	1.00

At the final row, the admitted correlation bound is approximately 0.4898, while the actual provider failures remain independent. The larger model share transforms those unchanged provider risks into greater aggregate exposure.

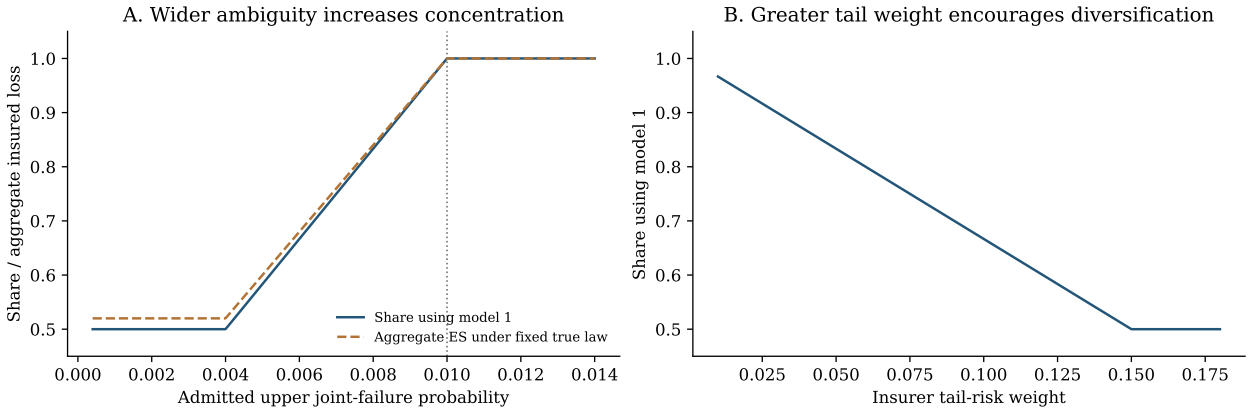


Figure 4: Architecture choices with fixed provider fees. Panel A varies the upper joint-failure probability at fixed true provider law and tail weight. Panel B fixes that upper bound at 0.006 and varies the tail weight; the willingness-to-pay condition remains satisfied throughout. Both panels keep marginal probabilities and total deployment fixed, and all firms remain insured. Inputs are illustrative.

The broad link from correlation ambiguity to concentration is established in Jiang et al. (2022). Here it appears through an insurance price difference and a specific tail-saturation threshold. The fixed-fee assumption is material, as the following audit demonstrates.

7.4 When provider prices adjust

The fixed-fee architecture result does not determine who will finance a control when model providers choose prices. Appendix D compares three specified markets. On an affine ES-cost branch, providers' combined profits fall even though market surplus rises. With a smooth portfolio cost, fees can instead rise enough to reduce customer surplus despite lower insurance quotes. Voluntary adoption and a fixed risk-bearing endowment do not eliminate this possibility; the latter introduces insurer rents and changes the participation boundary.

These are conditional incidence results, with global deviations checked in their stated domains. They do not establish a universal effect of safety on prices, a noncooperative investment equilibrium, or a justification for omitting customers from project finance. They explain why the market gain used in Sections 5 and 6 must be distinguished from the gain available to a particular payer.

8 System-wide insurance and resource accounting

"System-wide" refers here to a consolidated view of the shared liability, including its allocation between primary insurers, reinsurers, and policyholders. It does not mean universal coverage of every AI harm.

In the constant-returns mean–ES benchmark, identical insurers with the same ambiguity set cannot manufacture diversification by splitting a common large-pool liability into nonnegative quota shares. If $\sum_j x_j = x$, positive homogeneity gives

$$\sum_j \rho_u(x_j Y) = \sum_j x_j c(u) = x c(u) = \rho_u(xY). \tag{62}$$

This is a linear quota-share statement; it is not a theorem that reinsurance is generally useless. Differences in risk-bearing preferences, unrelated portfolios, actual external wealth, and information can support beneficial trades.

For example, let carrier j have a maximum funded exposure $\bar{x}_j$ and a per-unit reservation price $c_j(u) = p + \lambda_j[r_j(u) - p]$. A transparent allocation of total exposure x solves

$$\min_{\{x_j\}} \sum_j c_j(u) x_j \quad \text{subject to} \quad \sum_j x_j = x, \quad 0 \le x_j \le \bar{x}_j. \tag{63}$$

Absent other constraints, lower-cost capacity is used first. The homogeneous unlimited-capacity benchmark in the preceding sections is one special case, not a complete market equilibrium with endogenous balance sheets.

Cash accounting is distinct. If a primary insurer owes policyholders S, receives realized reinsurance recovery T, and the reinsurer owes T, consolidated net liability is $(S - T) + T = S$. Reinsurance receivables are not additional external assets. Let A_I and A_R denote disjoint external assets of the two entities, including retained premiums only once. Aggregate available funding is $A_I + A_R$, not $A_I + A_R + T$. Legal segregation and timing can make this aggregate necessary condition insufficient for individual payment. Wrong-way counterparty exposure requires modeling realized recoveries rather than using promised recoveries as cash.

The fixed-risk-tolerance extension in Section D.2 admits insurer rents above reservation compensation. Those rents enter participant surplus; they are not additional external assets beyond the premium receipts already included in cash accounts. Fixed risk tolerance changes the price response to book size without introducing a default model.

In the main model, prefunding of the maximum covered loss deliberately makes these constraints slack. Replacing prefunding by an expected-shortfall capital rule would require an explicit default model, customer valuation of impaired claims, and a treatment of tail losses beyond the capital buffer. No such equivalence is assumed here. A public backstop would likewise add an explicit contingent public liability; it would not remove uncertainty or create free resources.

9 Learning, Knightian uncertainty, and the limits of evidence

The model distinguishes three changes. Physical prevention changes the loss process. Verification makes a promised control contractible. A diagnostic experiment changes information and may reveal either good or bad news. Only the first two are bundled in the deterministic assurance technology.

Suppose, separately, that an audit can produce a favorable certificate with monetary continuation gain $\Delta > 0$ or an unfavorable result with zero gain relative to a no-audit baseline. Let the credible probability of a favorable result lie in $[\underline{\pi}, \overline{\pi}]$. Under an ex ante maxmin expected-money evaluation of these continuation gains, an audit costing C has value

$$V_A = \min_{\pi \in [\underline{\pi}, \overline{\pi}]} \{\pi\Delta - C\} = \underline{\pi}\Delta - C. \tag{64}$$

If $\underline{\pi} = 0$, favorable information can be highly valuable after it arrives while the costly audit has no positive guaranteed ex ante value. This calculation is conditional on the stated gain structure and commitment criterion. It is not a dynamically consistent updating theorem for expected shortfall. A full learning model must specify the experiment, the joint ambiguity set over signals and losses, and posterior updating.

The distinction prevents a common circular argument: knowing in advance that a purely informational audit will certainly prove a tighter bound would itself undermine the claimed pre-audit uncertainty. The physical intervention in (11) avoids that circularity by changing the system before making its reduced dependence verifiable.

No finite-dimensional ambiguity set can encompass all unforeseen failure modes. Moreover, if uncapped indemnities are evaluated over a class containing arbitrarily severe losses with a fixed positive probability, worst-case expected shortfall is infinite. Bounded insurance layers make the insurer's promised payout finite; they leave residual social harm outside the contract. This is why novelty of AI risk motivates careful uncertainty modeling but does not, by itself, justify any particular numerical ambiguity radius.

10 Illustration and empirical discipline

The following values are illustrative, not estimates: $p = 0.02$, $\alpha = 0.99$, $\lambda = 0.15$, $u_0 = 0.90$, $u_1 = 0.20$, $N = 100$, $w_i = 0.14$, and $F = 3$. One unit is the maximum insured loss per unit exposure mass.

Initially $r(u_0) = 1$ and $c(u_0) = 0.167$, so there is no insurance demand. Full assurance gives $r(u_1) = 0.412$ and $c(u_1) = 0.0788$. The saturation threshold is $u_* = 0.5$, reached at $e = 4/7 \simeq 0.5714$. Insurance demand appears only after $e_w \simeq 0.7026$. Each association gains 0.0612 under full assurance; total gains are 6.12. Thus $0.0612 < F = 3 < 6.12$, meeting Proposition 2. Equal procurement fees of 0.03 leave every association a positive net gain of 0.0312.

These are comparative statics of a constructed economy. The numbers neither estimate market size nor identify an AI catastrophe probability. A useful empirical design would require a defined insured layer, common-component exposure maps, verifiable changes to controls, policy prices and exclusions, and evidence on customer willingness to pay. Observations should be grouped by shared failure episode and model version; many users exposed to one update do not constitute many independent common-shock observations.

For the same illustration, the assurance benefit peaks at $\lambda \simeq 0.12245$, where total gains are 7.20. Net collective gains are positive only for $0.05102 < \lambda < 0.22959$. Thus the baseline $\lambda = 0.15$ lies inside a nontrivial financing window. At $\lambda = 0.30$, total benefit is only 0.24, below the project cost. Figure 5 displays both mechanisms.

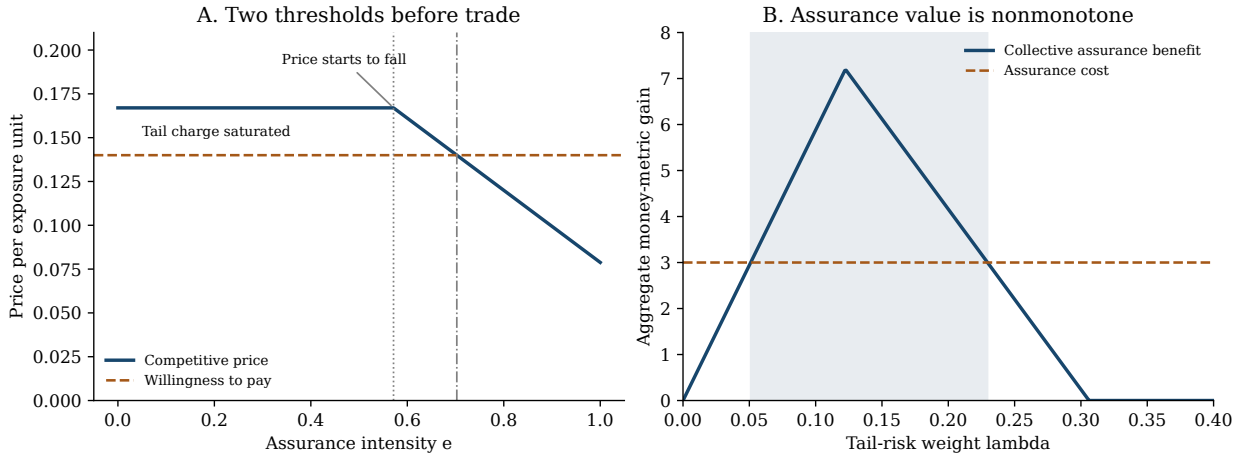


Figure 5: Illustrative thresholds and assurance benefits. Panel A holds $\lambda = 0.15$ fixed. Panel B varies λ for the same full-assurance project; the shaded interval satisfies (17). All inputs are chosen examples, not estimates.

The model yields a conditional contrast: an intervention that tightens a credible joint-dependence bound can alter quotes at unchanged marginal claim frequency; one that improves a benchmark without tightening that bound need not. Identification would require separating this channel from changes in loss severity, policy wording, insurer competition, and balance-sheet capacity. No such dataset or identification result is supplied in this draft.

11 Scope of the result

The paper separates two effects of greater tail concern. Increasing the weight on a fixed risk charge can reduce assurance value because coverage contracts. Concentrating the charge on increasingly extreme losses can remove its ability to reward a frequency reduction when the largest credible payout remains unchanged. With costly physical control, this can require increasing effort before a discontinuous switch to no investment, while insurance remains active. The general abandonment argument requires positive residual common risk at every finite effort and coercive cost; finite-cost elimination is a countercase. A finite-pool audit identifies a separate limit-order restriction.

Generating market value is distinct from collecting it. The private-benefit analysis gives a funding bound under efficient allocation, dominant-strategy truth telling, and participation relative to no project. The

architecture and pricing extensions show how gains can shift across customers, model providers, and insurers. The results do not establish an optimal risk criterion, insurer capital rule, universal form of AI insurance, or total social valuation of safety.

The main unresolved tasks are economic: establish the increment beyond known prevention, risk-measure, and public-project results; validate a control that changes dependence while preserving marginal losses; model private information about the risk process; and discipline magnitudes. The costly-control benchmark specifies a technology and solves its collective choice, but does not establish that this technology describes AI safety engineering or that its gains can be collected. Unknown control effectiveness, uncertain certification, and uninsured social benefits remain outside that choice. These limitations delimit what the present market analysis establishes.

A Finite pools and perturbation checks

Write $Z_n = n^{-1}\sum_{i=1}^n B_i$ and $Y_{n,\theta} = RB+(1-R)Z_n$. The limiting variable can be coupled as $Y_\theta = RB+(1-R)p$. We do not infer a limit by exchanging a supremum with convergence without a uniform bound.

Proposition 14 (Uniform actuarial approximation)**.** *Let $r_n(u) = \sup_{0\le\theta\le u} \mathrm{ES}_\alpha(Y_{n,\theta})$. Its worst case is $\theta = u$, and*

$$0 \le r_n(u) - r(u) \le \frac{1-u}{\tau}\sqrt{\frac{p(1-p)}{n}} \le \frac{1}{\tau}\sqrt{\frac{p(1-p)}{n}}. \tag{65}$$

For fixed n, $r_n(u)$ is nondecreasing and concave in u. If $p > \tau$ and $p^n < \tau$, it equals one exactly when

$$u \ge \frac{\tau - p^n}{p - p^n}. \tag{66}$$

Proof. The Bernoulli variable B dominates Z_n in convex order: for any convex f on $[0,1]$, $f(z) \le (1-z)f(0) + zf(1)$, and $\mathbb{E}Z_n = p$. Thus the stop-loss expectation for the mixture $Y_{n,\theta}$ is nondecreasing in θ at every threshold. Taking the infimum in (4) preserves that order, giving the worst-case endpoint. The same representation is an infimum of affine functions of θ, so it is concave.

Jensen's inequality gives $p \le_{cx} Z_n$, hence $Y_\theta \le_{cx} Y_{n,\theta}$ and $r_n(u) \ge r(u)$. From (4), expected shortfall obeys $|\mathrm{ES}_\alpha(X) - \mathrm{ES}_\alpha(X')| \le \mathbb{E}|X - X'|/\tau$ for coupled integrable variables. The coupling above gives

$$\mathbb{E}|Y_{n,u} - Y_u| = (1-u)\mathbb{E}|Z_n - p| \le (1-u)\sqrt{p(1-p)/n},$$

proving the uniform bound. Finally, for a bounded variable with maximum one, expected shortfall is one precisely when the mass at one is at least τ. That mass is $up + (1-u)p^n$, giving the stated condition. □

This validates the large-pool actuarial approximation. It does not solve a finite-market equilibrium in which the number of insured clients changes the diversification available to each carrier. The market propositions use the limiting per-unit quote, and the fixed-n concavity observation alone does not remove that restriction.

For general nearby laws, suppose every admitted $\widetilde{Y}_\theta$ can be coupled to Y_θ with $\mathbb{E}|\widetilde{Y}_\theta - Y_\theta| \le \varepsilon$, uniformly in θ, and the same means p. Then the two robust ES charges differ by at most ε/τ, and their mean–ES prices differ by at most $\delta = \lambda\varepsilon/\tau$. Each endpoint incremental benefit changes by at most 2δ, and total endpoint gains change by at most $2N\delta$. Thus the sign of $B(1) - F$ survives perturbations smaller than its slack. This does not establish exact plateaus, endpoint optimality, or equilibrium persistence for arbitrary perturbed laws. In particular, the error bound deteriorates as the tail probability τ becomes small.

B Risk-criterion sensitivity

A smooth alternative is the entropic reservation value for a unit loss, with parameter $\gamma > 0$:

$$c_\gamma(u) = \frac{1}{\gamma} \log \left[(1-u)e^{\gamma p} + u(1-p+pe^{\gamma}) \right]. \tag{67}$$

The term $D = 1 - p + pe^{\gamma} - e^{\gamma p}$ is strictly positive by convexity of the exponential. Thus $c'_\gamma(u) = D/[\gamma(e^{\gamma p} + uD)] > 0$ and $c''_\gamma(u) < 0$. There is no exact saturation interval. With the same affine assurance technology and a unit quote defined by (67), customer benefits are nevertheless convex in assurance, so the endpoint investment argument survives.

This is a fixed-unit price comparison. Entropic valuation is not positively homogeneous, so it does not preserve the constant-returns insurance supply model or the quota-share identity in (62). A market equilibrium must separately specify risk-bearing supply and book size; Section D.2 supplies proportional and fixed-risk-tolerance benchmarks with the appropriate profit accounting. The comparison establishes that the exact initial dead zone is criterion-dependent, while convex gains from a linear reduction in mixture dependence are more general. It does not establish robustness of the entire equilibrium to changing preferences.

C Reproducible numerical verification

The accompanying verification program uses no empirical observations. With random seed 20261101 it compares (5) with an independent linear-programming dual calculation of expected shortfall for 500 draws of (p, τ, u) spanning both saturation regimes. The maximum absolute discrepancy is 1.11×10^{-16}. It also computes exact binomial-mixture tail averages for 16 parameter/pool-size cases, checks (65), and checks monotonicity and concavity on three fixed-pool grids. These are numerical checks, not replacements for proofs.

For the displayed example, a 1001-point investment grid confirms that a unilateral contributor prefers zero while the collective objective prefers full assurance. An independent grid in λ locates the peak within 0.001 of (15). The figure and all numerical values are generated by the same reproducible script. The verification file records inputs, errors, and the finite-pool calculations; it contains no fitted parameters or simulated claims presented as data.

A separate extension program checks continuous coverage. It compares closed-form surplus with direct scalar maximization in 500 random linear-demand economies, including boundary optima. For 200 nonlinear-demand cases it compares the surplus difference with numerical integration of equilibrium coverage, and the analytical derivative with a central difference. It checks the active-market peak, the fixed-control ambiguity comparison, and convexity of optimized assurance gains in the displayed example. The accompanying output records numerical errors and all illustrative parameters. These checks supplement the proofs and make no claim about empirical fit.

The private-benefit program independently optimizes all payments in 16 finite-grid economies. Each linear program fixes efficient project selection and imposes every dominant-strategy reporting constraint, ex post participation, and a uniform funding bound. Payments are unrestricted in sign; the critical-payment form is not imposed. Discrete grids can permit higher charges than a continuous type interval. The computed optima agree with the grid-specific threshold bound, and their gap from (44) is bounded by N times the grid spacing. A further 10,000 random continuous-type comparisons check the proposed payments against alternative

reports, participation, and the funding bound. The largest numerical violation is below 3×10^{-16}. These computations are diagnostics for the analytical proof, not evidence on actual AI insurance markets.

The bundle audit instead optimizes total payments using $v_i q_i - P_i$ as utility, imposing efficient insurance and project allocations and the outside option $O(v_i)$. Twenty such linear programs are paired with twenty project-fee programs on the same valuation grids. These finite grids are relaxations of the continuous domain, not exact replicas. If a grid contains both endpoints and has maximum spacing Δv, its bundled optimum $S^*_{\mathrm{B,grid}}$ satisfies

$$0 \leq S^* - S^*_{\mathrm{B,grid}} \leq N\Delta v. \tag{68}$$

The upper bound on the grid optimum follows by restricting the continuous mechanism to grid reports. For the reverse bound, repeat the all-high-profile argument in Proposition 12. Replacing a continuous coverage threshold by its first winning grid type increases the maximum collectible payment by at most Δv per association. In the zero-funding regimes both optima are zero. This proves (68) and explains why apparent gains from discrete screening need not survive a continuous type domain.

All twenty grid optima satisfy this bound, with nested grids weakly tightening it. Constraint residuals are below 3×10^{-17}. A further 10,000 continuous valuation profiles and alternative reports check the constructed joint contract, including the coverage assigned to a false report. The largest incentive, participation, or funding-bound discrepancy is below 2×10^{-15}. The accompanying files record every parameter and optimizer result.

The architecture program checks the four-state expected-shortfall formula against 500 independent dual linear programs, with maximum error below 9×10^{-16}. It checks the worst model on 31-point dependence grids. In 500 economies with uniform or power-distributed productivity gains, direct surplus maximization agrees with the analytical share to within 5×10^{-8}, and objective values to within 2×10^{-16}. Demand clearing and zero-profit conditions hold to numerical tolerance. Ten thousand alternative insurer books yield no positive risk-adjusted profit beyond rounding error. Forty discretized allocation linear programs, allowing fractional assignment at a boundary, agree with the continuum share to within one population-grid cell. The program also verifies both comparative-static signs and the actual-law tail values in the table. These checks use no empirical observations.

The provider-pricing audit independently maximizes continuation surplus in 500 cases; the largest share discrepancy is below 3×10^{-8}. It globally optimizes both providers' deviations at 404 stationary candidates, splitting each price domain at every demand kink and including all boundaries. The deviation-gain formula agrees within 3×10^{-16}. In 200 cases strictly above the threshold, a profitable deviation is verified; within the proved existence region no gain exceeds 2×10^{-16}. Eighty further comparisons each challenge a global best response with 4,001 price points. One hundred finite-difference checks verify the signs and formulas for profit and surplus incidence. These tests audit the proof below; price-grid searches alone would not prove equilibrium nonexistence.

D Strategic provider prices and the incidence of safety

These extensions audit who receives the gains from a dependence control. They qualify private funding incentives while retaining the restrictions of each specified market.

D.1 Provider prices and the incidence of safety gains

Now let b be a *gross* productivity advantage, uniform on $[0, B]$, rather than an advantage net of fixed model fees. Two zero-cost providers simultaneously choose nonnegative fees f_1, f_2. Firms then choose models and buy insurance at competitive joint-book quotes. Each firm must use one of the two models; total deployment and full insurance remain fixed. The continuation allocation is unique, but the providers are strategic. Appendix E derives demand for every fee pair and checks all unilateral price changes.

Proposition 15 (A limited provider-pricing extension)**.** *Put* $\delta = \lambda(1 - \min\{\bar{t}/\tau, 1\})$ *and* $d = \delta/B$*. A pure-strategy provider-price equilibrium exists if and only if*

$$0 \le d \le d_* := 5 - \frac{3}{2}\sqrt{11} \simeq 0.02506. \tag{69}$$

When it exists, it is unique and satisfies

$$f_1^* = \frac{2B - \delta}{3}, \qquad f_2^* = \frac{B + \delta}{3}, \qquad s_P^* = \frac{2 - \delta/B}{3} > \frac{1}{2}. \tag{70}$$

At fixed p and λ, a decrease in $\bar{t} < \tau$ increases δ while keeping c_H fixed. Along the interior of this equilibrium region, combined provider profits Π_P and risk-adjusted market surplus W satisfy

$$\frac{d\Pi_P}{d\delta} = \frac{-2 + 4d}{9} < 0, \qquad \frac{dW}{d\delta} = \frac{2 + 5d}{9} > 0. \tag{71}$$

The fee difference falls by two thirds of a rise in δ. Thus $\partial s_P^*/\partial\delta = -1/(3B)$, one third of the response holding fees fixed on the same unbalanced-demand branch. Wider dependence ambiguity still increases concentration within this region. At the ES saturation threshold, $\delta = 0$ and the model-1 share is $2/3$, rather than the complete concentration obtained with zero fixed fees.

For the incidence statement, provider fees are transfers. In units excluding a common constant benefit,

$$W = B(s_P^* - (s_P^*)^2/2) - C(s_P^*, 1 - s_P^*), \qquad \Pi_P = B\left[(s_P^*)^2 + (1 - s_P^*)^2\right].$$

If a verified, marginal-preserving control raises δ between two points in the equilibrium region, its gross market gain is positive but its effect on combined vendor profits is negative. A sufficiently inexpensive control has positive net market surplus. A coalition funded only by these vendors cannot cover its positive cost and compensate both vendors for their profit changes. Transfers from customers or other beneficiaries could alter that financing conclusion; they are not supplied by this result.

The existence bound is restrictive. The nonzero-credit examples in Figure 4 need not belong to it. Beyond d_*, the displayed stationary prices are not an equilibrium and cannot support comparative-static or welfare claims. No mixed-strategy equilibrium, price cycle, or empirical instability is inferred. The audit identifies a limit of the fixed-fee benchmark, not a universal effect of dependence uncertainty.

D.2 Risk-bearing capacity, adoption, and financing incidence

The vendor-profit decline in Proposition 15 occurs on an affine segment of the ES portfolio cost. Smooth costs can also change how strongly customers respond to a vendor fee. To distinguish that response from an

assumption about insurance supply, we compare proportional and fixed risk-bearing capacity within the same voluntary-adoption environment.

Retain the four-state provider loss law and put $a = p - \bar{t} \in [0, p - p^2]$. Customers choose between two symmetrically differentiated models and a no-adoption option. Their money-metric utility from insured model j is $v + \epsilon_j - f_j - \pi_j$, and from non-adoption is ϵ_0. The three tastes are independent type-I extreme value with scale $\sigma > 0$, independent of losses. Uninsured use replaces the premium by the common certainty-equivalent loss w. In the domains below, insurance is strictly preferred conditional on use, while adoption is voluntary. This demand specification is a tractable benchmark, not an estimate from deployment data.

Let $q_1 + q_2 = Q$ and $s = q_1/Q$. Under proportional capacity, a book of mass Q shares its liability among a mass Q of CARA risk bearers with coefficient $\gamma > 0$, each externally prefunded for the unit layer. Worst-model reservation compensation is $C(Qs, Q(1-s), a) = Qc_\gamma(s, a)$, where

$$c_\gamma(s, a) = \frac{1}{\gamma} \log[M_0 - aK_\gamma(s)], \quad M_0 = 1 - p + pe^\gamma, \quad K_\gamma(s) = e^\gamma + 1 - e^{\gamma s} - e^{\gamma(1-s)}. \tag{72}$$

The ambiguity interval is $t \in [p^2, p - a]$, without a prior. Independent replacement of the provider-pair draw implements the reduction while preserving marginal failure probabilities. Appendix F derives the cost and explains why this supply assumption makes it homogeneous. Supporting quotes then exhaust reservation compensation: $\pi \cdot q = C$.

Now fix the mass of risk bearers at R, with aggregate absolute risk aversion $\eta = \gamma/R$. Their fixed external assets suffice to prefund the entire covered layer. Equal sharing gives

$$C_\eta(q, a) = \frac{1}{\eta} \log\left[(p - a)e^{\eta Q} + ae^{\eta q_1} + ae^{\eta q_2} + 1 - p - a\right]. \tag{73}$$

This cost is convex but not homogeneous. Competitive quotes are its gradient, and risk bearers receive certainty-equivalent rents

$$\pi = \nabla_q C_\eta, \qquad \Pi_I = \pi \cdot q - C_\eta \geq 0.$$

Rents arise from a fixed endowment of risk tolerance, not insurer market power. Entry of additional risk bearers is not free in this benchmark. Fixed risk tolerance also differs from cash capital: CARA preferences do not by themselves make wealth determine absolute risk aversion. Both supply specifications retain prefunding and exclude default.

Vendors simultaneously choose nonnegative fees, anticipating customer choices and competitive insurance supply. Write $C(Q, s, a) = C(Qs, Q(1-s), a)$ for either cost. At a symmetric allocation, let

$$A = \frac{\sigma}{Q(1-Q)} + C_{QQ}, \qquad K = \frac{C_{ss}}{Q}.$$

The own-quantity response to a vendor fee is $-1/(4A) - Q/(4\sigma + K)$. The first term measures exit and the second substitution across providers. Therefore a symmetric stationary fee satisfies

$$f = \frac{2}{1/(QA) + 4/(4\sigma + K)}, \qquad f + C_Q = v + \sigma \log \frac{2(1-Q)}{Q}. \tag{74}$$

A dependence reduction can dampen provider substitution by increasing K, encouraging higher fees. With fixed risk tolerance it also changes the insurance-price feedback through total adoption, C_{QQ}. The second channel is absent under proportional capacity. Appendix H gives a common incidence identity and separates these first-order conditions from global equilibrium.

Proposition 16 (Fixed risk tolerance and a participation boundary)**.** *Fix $0 < p < 1/2$, $\sigma > 0$, $v \in \mathbb{R}$, and $w > p$. For sufficiently small positive η, uniformly over $a \in [0, p - p^2]$, the fixed-risk-tolerance economy has a unique pure vendor-price equilibrium. It is symmetric, with voluntary insured adoption and positive insurer rents. Let $Q^0 \in (0, 1)$ denote adoption in its risk-neutral limit:*

$$f^0 = \frac{2\sigma}{2 - Q^0}, \qquad \frac{Q^0}{1 - Q^0} = 2 \exp\left(\frac{v - f^0 - p}{\sigma}\right).$$

With $D = 4 - 2Q^0 - (Q^0)^2$, total customer price $P = f + \pi$ satisfies

$$P_a = \eta \frac{Q^0(3Q^0 - 2)(2 - Q^0)}{2D} + O(\eta^2), \tag{75}$$

uniformly in a. If $Q^0 > 2/3$, total price rises and adoption and customer surplus fall for all sufficiently small positive η. If $Q^0 < 2/3$, these signs reverse. At $Q^0 = 2/3$ the leading price response vanishes. In every case, sufficiently small positive η gives

$$\pi_a < 0, \quad f_a > 0, \quad (\Pi_P)_a > 0, \quad (\Pi_I)_a < 0, \quad W_a > 0, \qquad W = CS + \Pi_P + \Pi_I.$$

Here W is risk-adjusted market surplus before project cost.

The proof controls global asymmetric deviations and is uniform across the admissible control interval. It does not provide a numerical upper bound on η. The $2/3$ boundary describes the weak-risk limit; it is not an exact cutoff at arbitrary positive risk aversion. A known baseline away from the boundary gives strict signs throughout the control interval for sufficiently small η, hence the same signs for any finite dependence reduction within that interval.

The proportional-capacity comparison has a different boundary. Proposition 18 in Appendix G proves a local customer loss near $a = 0$ exactly when $Q_0 > 2 - \gamma/[2\sinh(\gamma/4)]$, with $w > c_\gamma(1, 0)$. As γ tends to zero, this threshold tends to zero. Thus customer losses do not require compulsory adoption or proportional capacity, but changing the supply specification changes their domain. At a risk-neutral baseline adoption of $1/2$, the weak-risk fixed-capacity case benefits customers, whereas the proportional-capacity case can harm them. At a baseline of 0.85, customer harm survives the fixed-capacity test.

Accounting for insurer rents is essential. At symmetry, $CS = -\sigma \log(1-Q)$, $\Pi_P = Qf$, and $\Pi_I = QC_Q - C$. De-risking lowers the payment needed to attract the endowed risk bearers, reducing their rents, while vendors gain from the changed demand response. An insurer-only financing calculation and a vendor-only calculation can therefore give opposite incentives for the same control. When customers lose, the combined vendor and insurer gain exceeds the total market gain. A coordinated coalition could then prefer a project whose cost exceeds its risk-adjusted market benefit. These are incidence and potential funding comparisons, not a noncooperative investment equilibrium.

The weak-risk coefficients use second moments of a bounded loss layer; they are a robustness check on financing incidence rather than a theorem about heavy-tailed severity. The central coverage result concerns how much assurance value the insurance market generates. This extension explains why that value, its distribution after prices adjust, and the gains available to particular payers need not coincide. It also identifies the supply and demand objects that a quantitative application would have to discipline, rather than treating a lower insurance quote as sufficient evidence of customer benefit.

E Provider-price continuation and global deviations

This appendix proves Proposition 15. Write $D = f_1 - f_2$. Conditional on fees, the strictly concave continuation objective is

$$B(s - s^2/2) - Ds - C(s, 1-s), \qquad 0 \le s \le 1.$$

Its maximizer is the competitive adoption allocation: the insurance quotes supporting C make the marginal firm indifferent whenever the share is interior. On the two sides of $1/2$, the derivative of the compensation function is respectively $-\delta$ and δ. Consequently, the continuation share for every fee pair is

$$S(D) = \begin{cases} 1, & D \le -\delta, \\ 1 - (D+\delta)/B, & -\delta < D < B/2 - \delta, \\ 1/2, & B/2 - \delta \le D \le B/2 + \delta, \\ 1 - (D-\delta)/B, & B/2 + \delta < D < B + \delta, \\ 0, & D \ge B + \delta. \end{cases} \tag{76}$$

All formulas agree at their endpoints. The balanced interval reflects a change in supporting insurance quotes, not rationing or an arbitrary consumer tie-breaking rule. Provider payoffs are $f_1 S(f_1 - f_2)$ and $f_2[1 - S(f_1 - f_2)]$.

Exhausting possible pure equilibria. At a full-concentration outcome the excluded provider can obtain positive demand at a positive fee. The only exception to this entry argument is the zero-credit, zero-fee corner for provider 1, where provider 1 itself can profitably increase its fee. Thus no corner is an equilibrium. At a balanced outcome with $\delta > 0$, provider 1 can raise its fee without losing customers unless $D = B/2 + \delta$; provider 2 can do the same unless $D = B/2 - \delta$. Both conditions cannot hold. When $\delta = 0$, a balanced outcome would require $D = B/2$, whereas both local optimality conditions require $f_1 = f_2 = B/2$, an impossibility.

At an interior unbalanced outcome a zero fee cannot be optimal: a small increase retains positive demand and earns positive profit. Both fees are therefore positive, and demand has slope $-1/B$ in each provider's own fee. The first-order conditions give $f_1 = Bs$ and $f_2 = B(1-s)$. On the branch $s < 1/2$, substitution into (76) would yield $s = (2+d)/3 > 1/2$, a contradiction. On the branch $s > 1/2$, it yields (70); membership in that branch requires $d < 1/2$. This is the only possible pure-equilibrium candidate.

Global rather than local optimality. For $0 \le d < 1/2$, fix provider 2's candidate fee f_2^*. Provider 1's profit is concave on each sloped-demand segment and increases linearly on the balanced interval. Its maximum over the branch with share above $1/2$ is the stationary candidate. Its maximum over the remaining branches is reached by raising its fee to

$$\widehat{f_1} = f_2^* + B/2 + \delta = \frac{5B + 8\delta}{6}, \qquad S(\widehat{f_1} - f_2^*) = \frac{1}{2}. \tag{77}$$

Beyond this price, profit is decreasing: the right derivative at the boundary is $1/2 - \widehat{f_1}/B < 0$ and becomes smaller. The price region with share one is already dominated by the stationary maximum on the adjacent segment. Hence provider 1 has no profitable deviation if and only if

$$\frac{(2B-\delta)^2}{9B} \ge \frac{5B + 8\delta}{12} \iff 4d^2 - 40d + 1 \ge 0. \tag{78}$$

Within $d < 1/2$ this is exactly $d \le d_*$.

Provider 2's only competing maximum is the upper-fee endpoint of its balanced interval, $f_1^* - B/2 + \delta$. If a lower-fee branch with majority share is feasible, its profit increases up to that interval and cannot improve on this endpoint. The candidate's profit exceeds the endpoint profit by

$$\frac{(B+\delta)^2}{9B} - \frac{1}{2}(f_1^* - B/2 + \delta) = \frac{B}{9}(d - 1/2)^2 \ge 0.$$

Thus provider 2 has no profitable deviation for any $d < 1/2$. Condition (78), together with the exhausted candidate list, proves existence, uniqueness, and nonexistence in the stated regions. At $d = d_*$ provider 1 is indifferent to the deviation in (77); the latter profile is balanced and is not a second Nash equilibrium.

Incidence and finance. On the equilibrium branch, $s = (2 - d)/3$ and both fee revenues are B times the square of the respective share. With c_H fixed,

$$\Pi_P = \frac{B}{9}(5 - 2d + 2d^2), \qquad W = \frac{B}{18}(8 + 4d + 5d^2) - c_H. \tag{79}$$

Differentiation gives (71). Customer surplus, excluding the same common constant, is $W - \Pi_P$ and has derivative $(4 + d)/9 > 0$. Provider 1 loses profit while provider 2 gains; the sum falls. For an intervention taking δ_0 to $\delta_1 > \delta_0$ within the equilibrium region, $\Delta\Pi_1 + \Delta\Pi_2 < 0$. Any vendor-only contributions satisfying individual participation, $\Delta\Pi_i - g_i \ge 0$, must therefore have $\sum_i g_i < 0$ and cannot finance a positive project cost. This remains true with unrestricted transfers between the vendors. It does not exclude customer payments.

The nonexistence result concerns this static game with an exact ES-induced kink, uniform gross productivity, zero production costs, compulsory adoption, and complete insurance. Smoother insurance costs, a participation margin, or other demand distributions require their own global analysis. Nothing here establishes that mixed equilibria fail to exist or that actual vendor prices cycle.

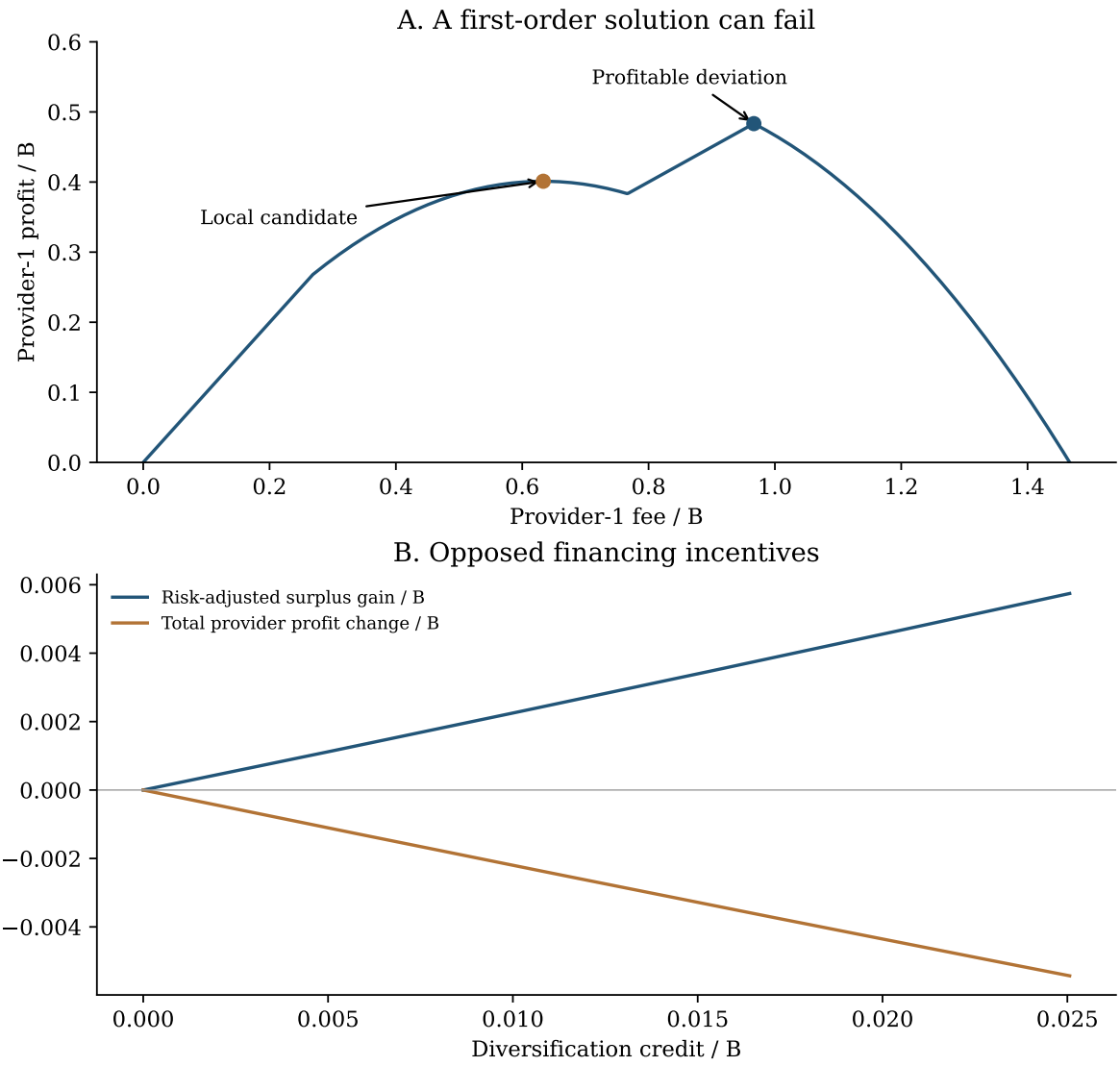


Figure 6: The global price audit and the incidence of a diversification credit. Panel A fixes $\delta/B = 0.1$ and the rival's stationary fee; the local candidate is defeated by a price increase to the balanced-share boundary. It is not an equilibrium. Panel B uses only $0 \le \delta/B \le d_*$ and holds c_H fixed. Both changes are relative to $\delta = 0$ and exclude project cost. Values are constructed, not empirical estimates.

F Smooth risk bearing and financing incidence

F.1 An incidence identity with general productivity demand

For a convex per-unit cost $c(s,a)$, write the constant-returns book compensation as $q\,c(x_1/q,a)$, where $q = x_1+x_2$ and a indexes a verified dependence reduction. This subsection fixes total adoption at one. Let $v(s)$ be the strictly decreasing gross productivity advantage of the marginal model-1 adopter, with the derivatives needed below. Competitive insurance quotes differ by c_s, so continuation satisfies $v(s) = f_1 - f_2 + c_s$. On a differentiable interior equilibrium branch define

$$h = -v'(s) + c_{ss}(s,a) > 0, \quad z = 2s-1, \quad A = s^2 + (1-s)^2.$$

The share response to the fee difference is $-1/h$. At an interior equilibrium with positive fees, vendor optimality therefore gives $f_1 = sh$ and $f_2 = (1-s)h$. Hence

$$G(s,a) := c_s(s,a) + (2s-1)h(s,a) - v(s) = 0, \qquad G_s = 3h + (2s-1)h_s. \tag{80}$$

Total vendor profit is hA. Define $W = J(s) - c(s,a)$ with $J' = v$. If $3h + zh_s \neq 0$, differentiation gives

$$s_a = -\frac{c_{sa} + zc_{ssa}}{3h + zh_s}, \qquad (\Pi_P)_a = Ac_{ssa} + (Ah_s + 2zh)s_a, \qquad W_a = zhs_a - c_a. \tag{81}$$

The profit identity uses $h_a = c_{ssa}$ at fixed s; the surplus identity uses $v - c_s = zh$. The term Ac_{ssa} captures a change in the slope of demand facing each provider; the remaining profit term captures adoption changes, including movement along an already curved cost. A reduction in the cost level alone does not sign either group's gain. These are identities along a regular, differentiable equilibrium branch. They do not prove that a stationary candidate is a global best response, or that an equilibrium exists for every demand or cost function. The following argument supplies that missing check for the smooth counterexample.

F.2 The insurance cost and the capacity assumption

For $q = x_1+x_2 > 0$, a mass q of identical risk bearers each receives premium C/q and pays loss $(x_1Z_1+x_2Z_2)/q$. Each has utility $-\exp(-\gamma w)$. Equality between insured and outside expected utility requires

$$C = q \sup_t \frac{1}{\gamma} \log \mathbb{E}_t \exp\left(\gamma \frac{x_1Z_1 + x_2Z_2}{q}\right).$$

The capital-provider mass is tied to insured exposure, and each bears at most the prefunded unit. There is no unrestricted free addition of capital to a fixed book. The construction changes supply relative to an insurer with fixed absolute risk aversion and fixed wealth; its aggregate risk tolerance grows with q.

For $s = x_1/q$, the moment generating function is

$$te^{\gamma} + (p-t)\left(e^{\gamma s} + e^{\gamma(1-s)}\right) + 1 - 2p + t.$$

Its derivative in t is

$$K_\gamma(s) = (e^{\gamma s} - 1)(e^{\gamma(1-s)} - 1) \geq 0.$$

The worst model is therefore the upper endpoint $t = p - a$, giving (72). For any admitted a, c_γ is symmetric and convex in s, being the log of a nonnegative weighted sum of exponentials of affine functions. Its

perspective $qc_\gamma(x_1/q, a)$ is convex and homogeneous. To see convexity directly, the portfolio share of a mixture of two books is a mixture of their shares with weights proportional to their masses; apply convexity of c_γ and multiply by total mass. The supporting premiums are

$$\pi_1 = c_\gamma + (1 - s)(c_\gamma)_s, \qquad \pi_2 = c_\gamma - s(c_\gamma)_s.$$

They earn zero risk-adjusted profit on the supplied book and no positive profit on any other book. At $a = 0$ the cost is the constant $c_0 = \gamma^{-1} \log M_0 \in (p, 1)$ and both quotes equal c_0. Continuity preserves positive quotes and full participation for all sufficiently small a when $w > c_0$.

The improved bound can represent a physical intervention. Starting from $t \in [p^2, p]$, retain the original provider-pair draw with probability r and otherwise replace it with independent Bernoulli(p) draws. Then $t' = p^2 + r(t - p^2)$ and both marginal probabilities stay p. Choosing $r = (p - a - p^2)/(p - p^2)$ produces the interval $[p^2, p - a]$. Verification concerns this intervention; the calculation does not assume an informational audit always delivers favorable news.

F.3 Compulsory-adoption comparison

Proposition 17 (A smooth reversal of vendor-profit incidence)**.** *Fix $0 < p < 1/2$, $\gamma > 0$, and $B > 0$, with uniform gross productivity on $[0, B]$. Maintain compulsory model adoption and full insurance, choosing $w > c_\gamma(1, 0)$. There exists $\varepsilon > 0$ such that, for $0 \le a < \varepsilon$, the provider-price game has a unique pure equilibrium with nonnegative insurance quotes. At $a = 0$, $s = 2/3$, $f_1 = 2B/3$, and $f_2 = B/3$. The right derivatives satisfy*

$$s_a < 0, \qquad (\Pi_P)_a > 0, \qquad W_a > 0. \tag{82}$$

For sufficiently small positive γ, customer surplus $W - \Pi_P$ has a negative right derivative.

The statement is local in a dependence improvement, with an explicit global-deviation proof in the appendix. In a constructed example with $p = 0.02$, $\gamma = 1$, $B = 0.12$, and $w = 0.2$, conservative bounds establish a unique pure equilibrium over the full admissible interval. As $\bar{t}$ falls from 0.02 to 0.0004, the model-1 share falls from 0.6667 to 0.6092. Combined vendor profit rises from 0.0667 to 0.0960, market surplus rises by 0.00513, and customer surplus falls by 0.02421. Both insurance quotes fall, while both model fees rise. All quantities are synthetic and exclude project cost.

F.4 Global existence near perfect admitted dependence

Here productivity is uniform, so $v(s) = B(1 - s)$ and $h = B + c_{ss}$. For a given a put

$$M_1 = \sup_s |c_s|, \qquad M_2 = \sup_s c_{ss}, \qquad M_3 = \sup_s |c_{sss}|, \qquad F = 2B + M_1 + M_2.$$

The following sufficient inequalities are convenient:

$$M_3 < 3B, \qquad FM_3 < 2B^2. \tag{83}$$

They hold near $a = 0$ because all three derivative bounds converge to zero. The equilibrium condition has derivative $G_s = 3h + (2s - 1)c_{sss} \ge 3B - M_3 > 0$. Symmetry gives $G(1/2) = -B/2$, while convexity gives $G(2/3) \ge 0$. Thus there is exactly one stationary share in $[1/2, 2/3]$.

At its candidate prices, each fee is at most $B + M_2$. A deviating provider with positive demand has a fee at most its rival's fee plus $B + M_1$, and hence at most F. On an interior-demand segment, its profit second derivative is one of

$$-\frac{2}{h} - \frac{f_1 c_{sss}}{h^3}, \qquad -\frac{2}{h} + \frac{f_2 c_{sss}}{h^3}.$$

Both are negative under (83). Any full-demand segment has profit increasing linearly in the fee; the derivative weakly drops when entering the interior-demand region. A zero-demand segment gives zero profit. The interior stationary price is consequently a global best response for each vendor.

No other pure equilibrium is omitted. At full concentration the excluded vendor can win positive demand at a positive fee, except at the zero-credit, zero-fee corner where the incumbent can profitably raise its fee. A zero fee with interior demand is not optimal. Every remaining equilibrium must therefore satisfy the unique stationary condition. This proves global existence and uniqueness near $a = 0$, rather than relying only on an implicit-function calculation.

F.5 Signs of the local incidence

Define

$$D_\gamma = \frac{e^{2\gamma/3} - e^{\gamma/3}}{M_0}, \quad T_\gamma = \frac{\gamma(e^{2\gamma/3} + e^{\gamma/3})}{M_0}, \quad L_\gamma = \frac{K_\gamma(2/3)}{\gamma M_0}.$$

At $a = 0$, $(c_s)_a = D_\gamma$, $(c_{ss})_a = T_\gamma$, $c_a = -L_\gamma$, and all pure share derivatives of c vanish. Since $s = 2/3$,

$$\begin{aligned} s_a &= -\frac{D_\gamma + T_\gamma/3}{3B} < 0, \\ (\Pi_P)_a &= \frac{13}{27} T_\gamma - \frac{2}{9} D_\gamma > 0, \\ W_a &= L_\gamma - \frac{1}{9} D_\gamma - \frac{1}{27} T_\gamma > 0. \end{aligned} \tag{84}$$

For the profit sign, $T_\gamma / D_\gamma = \gamma \coth(\gamma/6) \geq 6$. For the surplus sign, set $y = \gamma/6$ and use $\cosh(3y) - \cosh y = 4 \cosh y \, \sinh^2 y$ to obtain

$$W_a = \frac{2e^{\gamma/2} \cosh y}{9 y M_0} \left[6 \sinh^2 y - y \tanh y - 2y^2 \right] > 0.$$

The bracket is at least $3y^2$, because $\sinh y \geq y$ and $\tanh y \leq y$ for $y > 0$. Finally, as γ decreases to zero,

$$(\Pi_P)_a = \frac{8}{9}\gamma + O(\gamma^2), \qquad W_a = \frac{1}{9}\gamma + O(\gamma^2).$$

Thus $(W - \Pi_P)_a = -7\gamma/9 + O(\gamma^2) < 0$ for sufficiently small positive γ. This completes Proposition 17. By continuity the strict local signs persist on a nonzero interval of dependence improvements; the proof does not sign arbitrary large changes for every parameter choice.

F.6 A full-interval example and its independent checks

For numerical bounds one need not optimize derivatives over shares. Write $D_{\max} = e^\gamma - 1$ and $T_{\max} = e^\gamma + 1$. Because the moment generating function is at least one,

$$\begin{aligned} &M_1 \leq a D_{\max}, \qquad M_2 \leq a\gamma T_{\max}, \\ &M_3 \leq a\gamma^2 D_{\max} \left(1 + 3a T_{\max} + 2a^2 D_{\max}^2 \right). \end{aligned}$$

For $p = 0.02$, $\gamma = 1$, $B = 0.12$, and $a \le p - p^2 = 0.0196$, these upper bounds are 0.03368, 0.07288, and 0.04112. They yield $3B - M_3 > 0.3188$ and $2B^2 - FM_3 > 0.01455$, establishing global equilibrium existence and uniqueness over the entire illustrated interval.

The smallest supporting quote over all shares is at a pure-provider boundary:

$$\pi_{\min} = c_0 - \frac{a(e^{\gamma} - 1)}{M_0}.$$

This follows from $\partial_s \pi_1 = (1 - s)c_{ss} \ge 0$ and $\partial_s \pi_2 = -sc_{ss} \le 0$. In the example it remains above 0.00122, and both quotes are at most $c_0 = 0.03379 < w = 0.2$. Thus full insurance is verified even away from the equilibrium composition. Negative customer-surplus levels omit the common value of compulsory model use; only their changes are interpreted.

The code uses synthetic inputs only. It checks 500 direct four-state cost evaluations, with error below 3×10^{-15}; 120 one-sided local derivative calculations; and 100 nonlinear-demand conditional identities. Along 61 example points, both global best responses match candidate fees within 2×10^{-9}. Another 100 direct continuation optimizations agree within 4×10^{-8}. These audits supplement the proofs and do not establish global equilibria for general nonlinear demand.

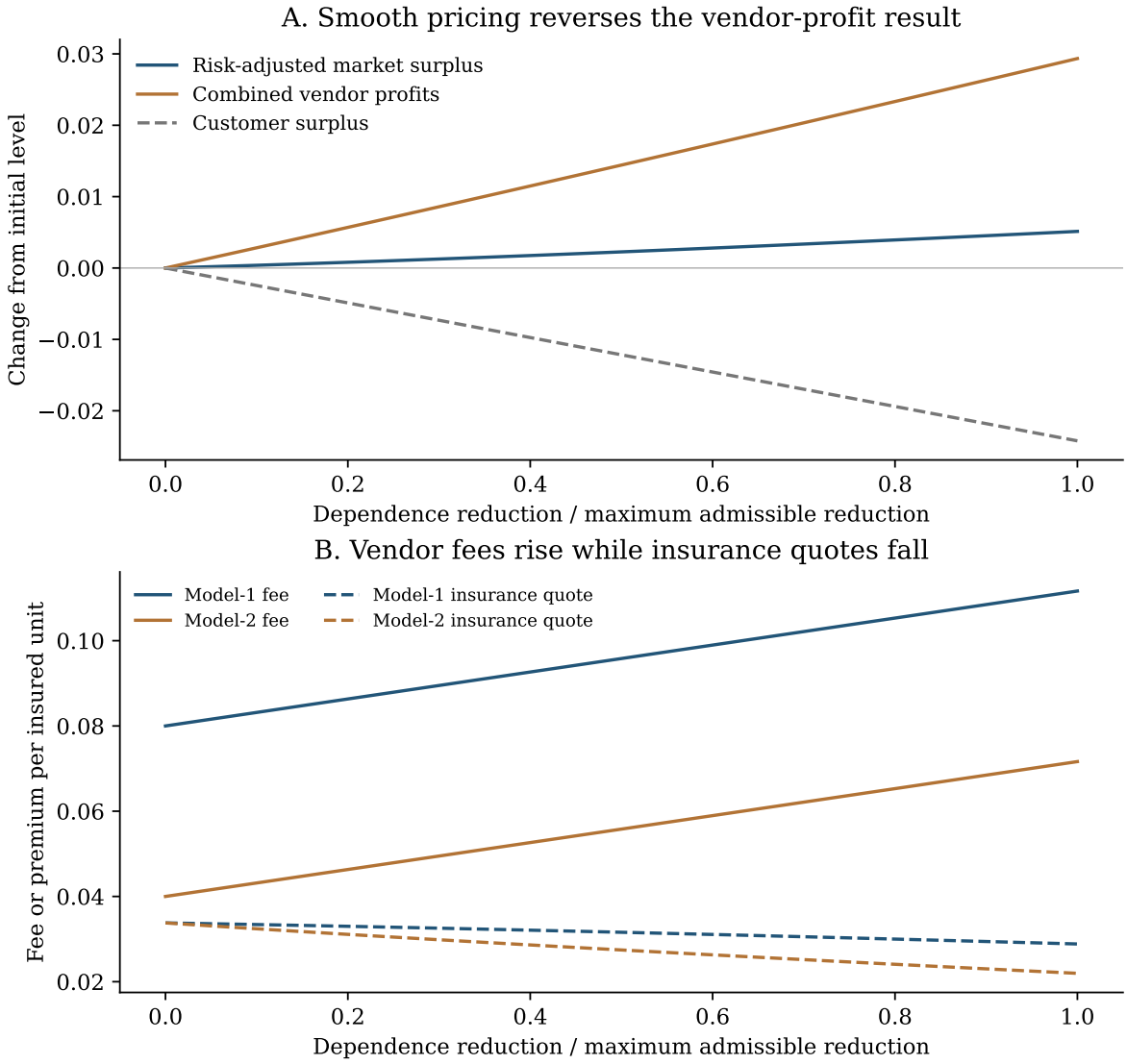


Figure 7: A constructed smooth-pricing counterexample. The horizontal axis is $a/(p - p^2)$, with $p = 0.02$, $\gamma = 1$, $B = 0.12$, and $w = 0.2$. Marginal loss probabilities and total deployment stay fixed. Panel A reports changes relative to the perfect-dependence underwriting bound, excluding project cost and a common model-use benefit. Panel B separates model fees from insurance quotes. The full-interval existence and participation bounds are proved in the text; these numbers are not estimates of AI insurance markets.

G Voluntary adoption and equilibrium incidence

Proposition 18 (Customer losses with voluntary adoption)**.** *Fix $0 < p < 1/2$, $\gamma, \sigma > 0$, $v \in \mathbb{R}$, and $w > c_0$. For all sufficiently small nonnegative a, the model has a unique pure provider-price equilibrium, with positive insurance quotes and voluntary insured adoption. The equilibrium is symmetric. At $a = 0$, its adoption mass $Q_0 \in (0, 1)$ is the unique solution of*

$$f_0 = \frac{2\sigma}{2 - Q_0}, \qquad \frac{Q_0}{1 - Q_0} = 2 \exp\left(\frac{v - f_0 - c_0}{\sigma}\right).$$

Define $b = (e^{\gamma/2} - 1)^2/(\gamma M_0) > 0$ and $r_\gamma = \gamma/[2 \sinh(\gamma/4)]$. The right derivative of the total customer price $P = f + c_\gamma(1/2, a)$ is

$$P_a = b \frac{r_\gamma^2 - (2 - Q_0)^2}{4 - 2Q_0 - Q_0^2}. \tag{85}$$

The insurance quote falls. Adoption and customer surplus fall if and only if

$$Q_0 > 2 - r_\gamma. \tag{86}$$

At equality their first derivatives vanish; below it they rise. Vendor profits and risk-adjusted market surplus have strictly positive right derivatives for every stated parameter configuration.

G.1 Choice, insurance, and continuation equilibrium

Consider a unit mass of customers and two zero-cost providers. Customer i obtains money-metric certainty equivalents

$$U_{ij} = v + \epsilon_{ij} - f_j - \pi_j \quad (j = 1, 2), \qquad U_{i0} = \epsilon_{i0},$$

where the three tastes are independent type-I extreme-value draws with scale $\sigma > 0$. Tastes are observed before choice and independent of the provider losses. The common mean v is unrestricted. The outside option uses no model and generates no covered loss. An uninsured use of model j instead gives $v + \epsilon_{ij} - f_j - w$, where $w > c_0$. This can be the certainty-equivalent loss of a customer with CARA preferences; identical marginals make it independent of model and dependence. All tastes and deterministic fees are measured in certainty-equivalent money. Since insurance quotes converge uniformly to c_0 as $a \downarrow 0$, insured use strictly dominates uninsured use at every fee pair for sufficiently small a. Adoption itself remains voluntary.

Let q_j be the mass choosing provider j, $Q = q_1 + q_2$, and $s = q_1/Q$. The insurer compensation is the same perspective $C(q_1, q_2, a) = Qc_\gamma(s, a)$ as in Appendix F, now allowing Q to vary. Supporting quotes satisfy

$$\pi_1 = c + (1 - s)c_s, \qquad \pi_2 = c - sc_s, \qquad q_1\pi_1 + q_2\pi_2 = C.$$

The last equality accounts for insurance payments exactly once. Capacity continues to scale with the book; a change in total adoption does not hold aggregate risk tolerance fixed.

The standard logit choice and money-metric surplus formulas are described by Train (2009, Sections 3.1 and 3.5). Here the quotes entering them are endogenous. A convenient way to establish a unique continuation is to maximize

$$\Phi(q; f, a) = vQ - f_1q_1 - f_2q_2 - Qc(s, a) - \sigma\left[q_1 \log q_1 + q_2 \log q_2 + (1 - Q) \log(1 - Q)\right] \tag{87}$$

on $q_1, q_2 \geq 0$, $Q \leq 1$, with $0 \log 0 = 0$ and $C(0, 0, a) = 0$. Entropy is strictly concave and C is convex. Finite fees therefore give a unique interior maximizer. Its conditions are exactly

$$v - f_j - \pi_j = \sigma \log \frac{q_j}{1 - Q}.$$

They imply both logit demand and supporting insurance prices. In particular,

$$\sigma \log \frac{s}{1 - s} + f_1 - f_2 + c_s = 0.$$

With $h = \sigma/[s(1 - s)] + c_{ss}$, differentiation at arbitrary fees gives

$$\frac{\partial s}{\partial f_1} = -\frac{1}{h}, \qquad \frac{\partial Q}{\partial f_1} = -\frac{Q(1 - Q)s}{\sigma}, \qquad \frac{\partial q_1}{\partial f_1} = -Q\left[\frac{(1 - Q)s^2}{\sigma} + \frac{1}{h}\right]. \tag{88}$$

For the total-demand derivative, the composition terms cancel by the preceding share condition. A composition change and market exit are distinct margins.

At symmetric fees $s = 1/2$. Write $k = c_{ss}(1/2, a)$ and $c = c(1/2, a)$. A symmetric interior price optimum must satisfy

$$f = \left[\frac{1 - Q}{2\sigma} + \frac{2}{4\sigma + k}\right]^{-1}, \qquad \frac{Q}{1 - Q} = 2 \exp\left(\frac{v - f - c}{\sigma}\right). \tag{89}$$

For each a the fee expression is increasing in Q, whereas the fee consistent with the demand equation is decreasing. There is exactly one symmetric stationary candidate. Global optimality, including asymmetric deviations, still requires the next argument.

G.2 Global price equilibrium for a nonzero neighborhood

All statements in this subsection are for a fixed p, γ, σ, v, w satisfying Proposition 18. Uniform convergence of c and its derivatives on $[0, 1]$ lets us choose a nonzero right neighborhood of $a = 0$ on which both insurance quotes are positive, less than w, and $0 \leq k(s, a) = c_{ss}(s, a) \leq \sigma$.

First, all best responses lie in a common compact fee interval. From (88), the absolute semi-elasticity of provider 1's quantity is

$$m_1 = -\frac{\partial \log q_1}{\partial f_1} = \frac{(1 - Q)s}{\sigma} + \frac{1 - s}{\sigma + k(s, a)s(1 - s)} \geq \frac{1 - q_1}{5\sigma/4}.$$

The analogous inequality holds for provider 2. Positive insurance quotes and the outside option imply $q_j \leq \exp[(v - f_j)/\sigma]$, independently of the rival's fee. Put

$$R = \max\{0,\ v + \sigma \log 2,\ 5\sigma/2\} + \sigma.$$

For $f_j \geq R$, $q_j < 1/2$ and $f_j m_j > 1$, so the profit derivative $q_j(1 - f_j m_j)$ is negative. No best response lies above R. A zero fee is also not optimal because its profit derivative is $q_j > 0$. Thus every equilibrium lies in $(0, R)^2$.

At $a = 0$, quotes are the constant c_0 and the ordinary logit first-order condition is

$$F_j := 1 - \frac{f_j(1 - q_j)}{\sigma} = 0.$$

For any rival fee, $f_j(1-q_j)$ strictly increases with f_j, so the first-order condition defines the unique global best response. Any equilibrium must be symmetric: a higher fee would give a smaller quantity, whereas $f_j = \sigma/(1-q_j)$ would then require a smaller fee. The unique symmetric solution is $f_0 = 2\sigma/(2-Q_0)$ and the baseline demand equation in the proposition.

At this solution, put $q = Q_0/2 < 1/2$. The diagonal derivatives of (F_1, F_2) are $-1/\sigma$ and the off-diagonal derivatives are $q^2/[\sigma(1-q)]$. Both eigenvalues are strictly negative. The implicit-function theorem therefore yields a locally unique, smooth stationary price pair for all sufficiently small nonnegative a; symmetry makes its fees equal.

This stationary pair consists of global best responses. At the baseline, each provider has a unique global optimum on $[0, R]$ and a strictly negative own-profit second derivative there, $-q/\sigma$. Uniform continuity preserves the strictly positive profit gap outside a small neighborhood of that optimum when the rival fee and a are close to their baseline values. Within that neighborhood the second derivative remains negative, so the continued stationary fee is the sole maximizer. This excludes both nearby and distant deviations.

Finally, any sequence of additional equilibria with $a \downarrow 0$ has a convergent subsequence in $[0, R]^2$. By continuity its limit would be an equilibrium of the baseline game, hence the unique baseline pair. Local uniqueness excludes such a second branch. Existence and uniqueness thus hold on a nonzero right neighborhood. The argument does not assign a numerical radius to that neighborhood or establish equilibrium at every admissible dependence bound.

G.3 The price, adoption, and surplus derivatives

All derivatives in this subsection are right derivatives at $a = 0$, with p, γ, σ, v, w fixed. Write $Q = Q_0$, $f = f_0$, and define the insurance saving and curvature change

$$b = -c_a(1/2, 0) = \frac{(e^{\gamma/2} - 1)^2}{\gamma M_0} > 0, \qquad \kappa = c_{ssa}(1/2, 0) = \frac{2\gamma e^{\gamma/2}}{M_0} > 0.$$

Differentiate the two equations in (89). With $P = f + c$ and $H = f^2 Q(1-Q)/(2\sigma^2)$,

$$P_a = \frac{f^2\kappa/(8\sigma^2) - b}{1+H}, \qquad Q_a = -\frac{Q(1-Q)}{\sigma} P_a, \qquad f_a = P_a + b > 0. \tag{90}$$

The first identity also applies locally to any smooth symmetric convex cost with constant baseline $c(s, 0)$, saving $b > 0$, and curvature change $\kappa \geq 0$, whenever the equilibrium continues. Customer harm requires $\kappa/b > 2(2-Q)^2$. A level saving without a curvature change gives $P_a < 0$ and expands adoption.

For the specified exponential cost,

$$\frac{\kappa}{b} = \frac{\gamma^2}{2\sinh^2(\gamma/4)} = 2r_\gamma^2, \qquad r_\gamma = \frac{\gamma}{2\sinh(\gamma/4)} \in (0, 2).$$

Substitution gives (85). Because its denominator $D = 4 - 2Q - Q^2$ is positive for $Q \in (0, 1)$, customer surplus falls exactly when $Q > 2 - r_\gamma$.

The expected money-metric customer surplus relative to the outside option alone is

$$CS = \sigma \log\left[1 + 2e^{(v-P)/\sigma}\right] = -\sigma \log(1-Q).$$

The normalized entropy potential gives the same value. Total vendor profit is $\Pi_P = Qf$ and risk-adjusted market surplus is $W = CS + \Pi_P$. This is the gross value of chosen alternatives minus insurers' reservation compensation; it excludes project costs and harms outside the insured layer. Its derivative is equivalently $W_a = Qb + fQ_a$. In particular,

$$CS_a = -QP_a, \qquad (\Pi_P)_a = Q\left[b + \frac{Q}{2-Q}P_a\right], \qquad W_a = Q\left[b - \frac{2(1-Q)}{2-Q}P_a\right]. \tag{91}$$

Let $x = P_a/b$. Since $0 < r_\gamma < 2$,

$$-1 < x < \frac{Q(4-Q)}{D}.$$

The lower bound makes $(\Pi_P)_a > 0$. The upper bound makes $W_a > 0$, since

$$(2-Q)D - 2(1-Q)Q(4-Q) = 8(1-Q)^2 + Q^2(2-Q) > 0.$$

Thus, whenever customers lose, vendor gains exceed the market gain; suppliers' aggregate private benefit can exceed the surplus available for project costs. If a project of size a costs $ka + o(a)$, then $W_a < k < (\Pi_P)_a$ describes a local incentive conflict for a provider coalition able to coordinate and transfer money. It is not a noncooperative investment equilibrium or an implementation result.

G.4 Synthetic illustration and independent checks

For $p = 0.02$, $\gamma = 1$, $\sigma = 0.06$, choose $v = 0.0721995$ to give baseline adoption $Q_0 = 1/2$. Let $w > c_0 = 0.0337883$. The initial vendor fee is 0.08. The local insurance-price derivative is -0.406857, but the vendor-fee derivative is 0.653590, so the total-price derivative is 0.246733. Adoption has derivative -1.028053, customer surplus -0.123366, vendor profit 0.244551, and market surplus 0.121184. These are derivatives per unit reduction in the admitted joint-failure bound, not finite percentage changes or estimates. At $\gamma = 1$ the customer-harm threshold is $Q_0 > 0.0206824$.

The verification program evaluates costs from the four state probabilities, checks one-sided derivatives in 240 synthetic economies, and includes both sides of the customer-incidence boundary. The largest scaled derivative discrepancy is below 3×10^{-8}. It independently maximizes the two-dimensional potential in 120 cases; the largest quantity discrepancy is below 7×10^{-7} and the objective discrepancy below 10^{-13}. Another 120 numerical price derivatives agree with (88) within 10^{-10}. Eighty price searches challenge asymmetric deviations from symmetric candidates close to the baseline, with no profit gain above 2×10^{-15}. These finite searches supplement the local global-equilibrium proof; they do not certify its neighborhood's numerical radius.

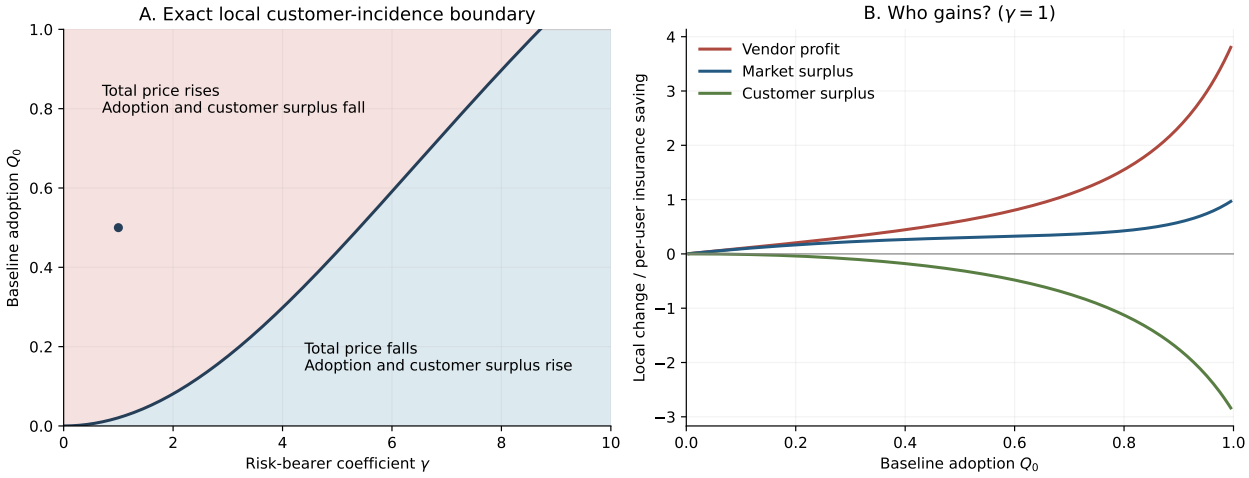


Figure 8: Local incidence with voluntary adoption. Panel A displays the exact sign boundary $Q_0 = 2 - r_\gamma$; the shaded regions concern a marginal dependence reduction at $a = 0$. Each point is a separately specified baseline economy, not a path holding mean valuation fixed as γ changes. Panel B uses $\gamma = 1$ and normalizes all derivatives by the per-user insurance saving b. Market and vendor gains remain positive while customer gains can be negative. These are theoretical comparative statics with synthetic inputs, not estimated AI-insurance outcomes.

H Fixed risk-bearing capacity and the incidence of safety

H.1 An endowment of risk bearers, rather than proportional entry

Keep the voluntary-choice environment of Appendix G. A mass $R > 0$ of identical price-taking risk bearers now has fixed CARA coefficient $\gamma > 0$ and fixed aggregate risk tolerance $R/\gamma = 1/\eta$. Each is externally prefunded for a maximum liability $1/R$, so aggregate funding of one suffices for the unit customer population. A risk bearer's chosen nonnegative insurance book has size at most $1/R$. This separates a cash-feasibility constraint from risk-bearing preferences; CARA preferences alone do not make greater initial wealth reduce absolute risk aversion.

For a given book $q = (q_1, q_2)$ with $Q = q_1 + q_2 \leq 1$, identical risk bearers optimally share it equally. The same upper endpoint $t = p - a$ maximizes exponential loss valuation for every nonnegative book. Convexity of the certainty equivalent and Jensen's inequality then give aggregate reservation compensation

$$C_\eta(q, a) = \frac{1}{\eta} \log \mathbb{E}_{p-a} e^{\eta(q_1 Z_1 + q_2 Z_2)}, \qquad \eta = \gamma/R.$$

Equal sharing attains this lower bound and respects each prefunding constraint. This is the explicit expression in (73). To verify the worst model, differentiate the moment generating function with respect to t:

$$\frac{\partial}{\partial t} \mathbb{E}_t e^{\eta(q_1 Z_1 + q_2 Z_2)} = (e^{\eta q_1} - 1)(e^{\eta q_2} - 1) \geq 0.$$

There is still no prior over t. A change in a uses the verified independent-replacement control already described in Appendix F.

For a competitive quote vector $\pi = \nabla_q C_\eta(q, a)$, convexity implies

$$\pi \cdot y - C_\eta(y, a) \leq \pi \cdot q - C_\eta(q, a) \quad \text{for every feasible book } y.$$

The same gradient supports each individual's optimum at book q/R. Consequently these are price-taking supply quotes, with aggregate certainty-equivalent rent

$$\Pi_I = \pi \cdot q - C_\eta(q, a) \geq 0. \tag{92}$$

The inequality follows by comparing with the empty book. At an interior supplied book it is strict for $\eta > 0$, because aggregate loss is nonconstant and the certainty equivalent is strictly convex along a scaling ray. Entry of additional risk bearers is not free in this benchmark. These rents are payments above reservation compensation, not extra prefunded assets and not compensation counted a second time. Utility-based equilibrium premium determination and aggregation of exponential risk tolerances are established methods; see Bühlmann (1980, Sections 2–6). The present market restricts the offered contracts and separately models vendor pricing.

H.2 A common continuation and pricing identity

Write $C(Q, s, a) = C(Qs, Q(1 - s), a)$ for either capacity specification. The continuation potential is

$$\Phi(q; f, a) = vQ - f \cdot q - C(q, a) - \sigma\{q_1 \log q_1 + q_2 \log q_2 + (1 - Q) \log(1 - Q)\}.$$

Convexity of C and strict concavity of entropy give a unique interior continuation at any finite fees. Its first-order conditions are

$$v - f_j - C_j(q, a) = \sigma \log \frac{q_j}{1 - Q}.$$

At symmetry $C_s = C_{Qs} = 0$. Put

$$D_0 = \frac{\sigma}{Q(1-Q)}, \qquad A = D_0 + C_{QQ}, \qquad K = \frac{C_{ss}}{Q}, \qquad H = 4\sigma + K.$$

Differentiating continuation with respect to one vendor's fee gives

$$Q_{f_j} = -\frac{1}{2A}, \qquad (q_j)_{f_j} = -\frac{1}{4A} - \frac{Q}{H}. \tag{93}$$

Thus a symmetric stationary fee and the demand equation satisfy

$$f = F(Q, a) := \frac{2}{1/(QA) + 4/H}, \qquad f + C_Q = v + \sigma \log \frac{2(1-Q)}{Q}. \tag{94}$$

The first denominator records exit; the second records substitution across providers. For a regular equilibrium branch,

$$Q_a = -\frac{F_a + C_{Qa}}{A + F_Q}, \qquad P_a = D_0 \frac{F_a + C_{Qa}}{A + F_Q}, \qquad P = f + C_Q. \tag{95}$$

The derivatives of F here are partial derivatives, holding Q fixed when differentiating in a. These identities alone do not establish a global price equilibrium.

Customer surplus remains $CS = -\sigma \log(1 - Q)$. Since homogeneity no longer eliminates insurer rents, the correct surplus accounting is

$$W = CS + \Pi_P + \Pi_I, \qquad \Pi_P = Qf, \qquad \Pi_I = QC_Q - C.$$

Equivalently, W is the gross money-metric value of choices less C. Its derivative and the two other directly useful derivatives are

$$CS_a = -QP_a, \qquad W_a = fQ_a - C_a, \qquad (\Pi_I)_a = QC_{QQ}Q_a + QC_{Qa} - C_a. \tag{96}$$

The optimized potential equals $CS + \Pi_I$, not CS alone. These distinctions are necessary to compare the fixed-endowment market with the proportional-capacity benchmark.

H.3 Uniform existence and uniqueness with weak risk aversion

Fix p, σ, v, w as in Proposition 16, and write $\nu = p(1-p)$. At $\eta = 0$, continuously extend C_η to pQ. This baseline is independent of $a \in [0, \nu]$, with insurance quotes p and zero insurer rent. For small positive η, quotes remain positive and less than w uniformly on all feasible books and all admissible a. Hence insured use strictly dominates uninsured use, including after any vendor fee deviation.

We first give a fee bound that is uniform in a. For positive η, the portfolio Hessian is

$$\nabla_q^2 C_\eta = \eta \, \mathrm{Cov}_{\mathrm{tilt}}(Z_1, Z_2).$$

The tilted covariance is positive semidefinite and has trace at most $1/2$, so $0 \preceq \nabla_q^2 C_\eta \preceq (\eta/2)I$. The entropy Hessian is

$$B_0 = \sigma \left\{ \operatorname{diag}(1/q_1, 1/q_2) + \frac{\mathbf{1}\mathbf{1}'}{1-Q} \right\} \succeq \sigma I.$$

Continuation has derivative $q_f = -(B_0 + \nabla_q^2 C_\eta)^{-1}$. The matrix bound implies the own-quantity semi-elasticity satisfies

$$-\frac{1}{q_j}\frac{\partial q_j}{\partial f_j} \geq \frac{1-q_j}{\sigma + \eta/2}.$$

For $\eta \leq \sigma/2$, the fee bound

$$R_f = \max\{0,\ v + \sigma \log 2,\ 5\sigma/2\} + \sigma$$

therefore works exactly as in Appendix G: positive quotes imply $q_j \leq e^{(v-f_j)/\sigma}$, and profit strictly decreases for $f_j \geq R_f$. Zero fees are not optimal. All equilibria lie in $(0, R_f)^2$.

At $\eta = 0$, each vendor has a unique global logit best response. The only equilibrium is symmetric, with adoption Q^0 and fee $2\sigma/(2-Q^0)$. Its normalized first-order-condition Jacobian has diagonal $-1/\sigma$ and off-diagonal $q^2/[\sigma(1-q)]$, where $q = Q^0/2 < 1/2$. It is nonsingular. The certainty equivalent, continuation, and profit derivatives converge smoothly to this common baseline uniformly in a, by bounded losses and compactness of $[0, v]$. The implicit-function theorem thus gives a unique nearby stationary pair uniformly for sufficiently small η.

At the baseline the own-profit second derivative at the optimum is strictly negative, and profit has a uniform positive gap away from any small neighborhood of that optimum. Both properties persist uniformly in a, so the continued stationary prices are global best responses. Any additional equilibrium as $\eta \downarrow 0$ would, by the common fee bound and compactness in a, have a subsequence converging to the unique baseline equilibrium. Local uniqueness rules it out. The resulting existence and uniqueness domain is uniform over the full admissible control interval, but the proof supplies no numerical value for its upper bound on η.

H.4 Expansion and the adoption boundary

Bounded losses permit the cumulant expansion, uniformly in quantities and a, with the derivatives used below:

$$C_\eta(Q, s, a) = pQ + \frac{\eta}{2}\left[vQ^2 - 2aQ^2 s(1-s)\right] + O(\eta^2). \tag{97}$$

At symmetry it gives

$$C_{Qa} = -\frac{\eta Q}{2} + O(\eta^2), \qquad C_{QQ} = \eta(v - a/2) + O(\eta^2), \qquad K = 2\eta a Q + O(\eta^2).$$

Differentiating (94) at fixed Q yields

$$F_a = \frac{\eta Q^2}{2-Q} + O(\eta^2), \qquad F_Q = \frac{2\sigma}{(2-Q)^2} + O(\eta).$$

Substitute into (95) and use $Q = Q^0 + O(\eta)$. With $D = 4 - 2Q^0 - (Q^0)^2$ and

$$L = \frac{Q^0(3Q^0 - 2)(2 - Q^0)}{2D},$$

the price and quantity derivatives are

$$P_a = \eta L + O(\eta^2), \qquad Q_a = -\eta \frac{Q^0(1-Q^0)}{\sigma} L + O(\eta^2), \qquad \pi_a = -\eta \frac{Q^0}{2} + O(\eta^2). \tag{98}$$

These are total equilibrium derivatives. The fee derivative is $\eta(Q^0)^2(3-2Q^0)/D + O(\eta^2) > 0$ for sufficiently small η. The leading premium decrease includes the fact that its feedback through Q_a is only second order.

Writing $q = Q^0$ just in the next display, the accounting identities give

$$\begin{aligned} CS_a &= -\eta q L + O(\eta^2), \\ (\Pi_P)_a &= \eta \frac{q^2(2-2q+q^2)}{D} + O(\eta^2), \\ (\Pi_I)_a &= -\eta \frac{q^2}{4} + O(\eta^2), \\ W_a &= \eta \frac{q^2[11(1-q)^2+1]}{4D} + O(\eta^2). \end{aligned} \tag{99}$$

The vendor and market coefficients are positive for every $q \in (0,1)$; the insurer coefficient is negative. The customer coefficient changes sign at $q = 2/3$. All remainders are uniform in $a \in [0, \nu]$ for fixed p, σ, ν, w. Strict derivative signs therefore hold throughout that interval for sufficiently small positive η whenever $Q^0 \neq 2/3$, and integrate to the corresponding finite-control comparisons. At $Q^0 = 2/3$, the leading price and customer coefficients vanish; higher-order terms must determine the response. The theorem does not assert an exact $2/3$ cutoff at an arbitrary positive η.

Insurance loses rent because de-risking reduces the premium needed to attract the same endowed risk bearers. Vendors gain because it changes the demand response relevant to their fees. For a coalition including both vendor and insurer gains, the leading private gain is

$$(\Pi_P + \Pi_I)_a = \eta \frac{q^2(4-6q+5q^2)}{4D} + O(\eta^2) > 0.$$

It exceeds W_a exactly when customers lose, to leading order. These comparisons concern potential coalition transfers before project cost; they do not specify an investment game. The sum of all participants' gains remains W_a.

H.5 Synthetic magnitudes and verification

Table 1 reports coefficients after dividing each derivative by η and taking the limit. The two baseline adoption levels are separate constructed economies. At $Q^0 = 1/2$, insurance savings more than offset fee increases and customers gain. At $Q^0 = 0.85$, fees more than absorb the insurance saving and customers lose. Insurer rents fall in both cases.

The verification program uses direct four-state exponential expectations and tilted moments. In 180 synthetic economies, exact analytical derivatives agree with independently resolved central differences within 10^{-11}. One hundred direct two-dimensional optimizations check continuation quantities to within 5×10^{-7} and objective values within 10^{-13}; one hundred numerical fee derivatives agree within 2×10^{-10}. Three thousand alternative insurer books give no positive profit deviation at supporting quotes. One hundred grids check the worst joint law. In 160 economies, halving η at three successive scales reduces the normalized error

Q^0	Total price	Customer surplus	Vendor profit	Insurer rent	Market surplus
0.50	−0.06818	0.03409	0.11364	−0.06250	0.08523
0.85	0.17040	−0.14484	0.46831	−0.18063	0.14284

Table 1: Leading derivatives with fixed risk tolerance. Each entry is $\lim_{\eta \downarrow 0}(\partial_a X)/\eta$. They are theoretical coefficients, not estimated outcomes or finite project gains.

from the leading-order coefficients by a factor no greater than 0.505, consistent with the proved expansion. Sixty numerical price searches find no vendor gain above 5×10^{-16}. These finite calculations do not certify the uniform existence radius or replace the proof.

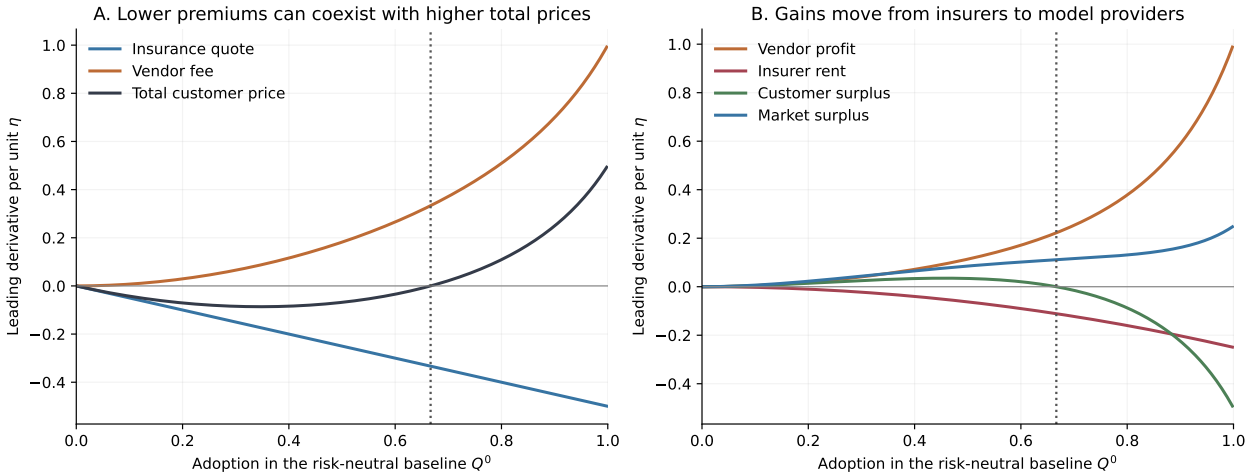


Figure 9: Fixed risk-bearing capacity changes the incidence boundary. Curves are leading coefficients of equilibrium derivatives with respect to a, per unit η, as $\eta \downarrow 0$. Panel A separates the insurance quote, vendor fee, and their sum. Panel B includes insurer rents in the accounting. The dotted line is $Q^0 = 2/3$, the leading-order customer-incidence boundary. This is not an estimated adoption threshold or an exact cutoff at arbitrary risk aversion.

I Tail focus, costly control, and finite pools

I.1 What the bounded-loss result does and does not require

Proposition 7 allows arbitrary nonempty ambiguity sets of bounded laws with a common mean. Neither a single worst law at all confidence levels nor compactness of the sets is needed. To see continuity explicitly, for any $\alpha, \beta \le 1 - \varepsilon$ the infimum representation with threshold $z \in [0, 1]$ gives

$$|\operatorname{ES}_\alpha^P(X) - \operatorname{ES}_\beta^P(X)| \le \frac{|\alpha - \beta|}{\varepsilon^2}.$$

The same bound holds for the supremum over P. For the extreme-tail limit, choose any $c < b_k$ and a law in $\mathcal{P}_k$ whose essential supremum exceeds c. Its ES eventually exceeds c, so $\liminf_{\alpha \uparrow 1} R_k(\alpha) \ge c$. Taking $c \uparrow b_k$ proves the lower bound; the reverse bound is immediate.

The relevant endpoint is the supremum of essential suprema over admitted laws. A small but positive admitted probability of a common maximum payout can therefore be decisive in the extreme tail. This is compatible with a control reducing risk under the actual law. It is also possible without ambiguity: singleton law sets suffice. Ambiguity matters by determining which endpoints and residual probabilities remain credible. Unknown severity outside the specified insured layer is not covered by the theorem.

The endpoint conclusion extends beyond the ES step weights. Let ϕ_m be nonnegative normalized weights on loss quantiles, with

$$\int_0^{1-\varepsilon} \phi_m(u)\, du \longrightarrow 0 \quad \text{for every } \varepsilon > 0.$$

For any bounded law, its weighted quantile average tends to its essential supremum. The same approximately maximizing-law argument gives convergence of the robust supremum to b_k. Thus equal robust endpoints again give vanishing differences in insurance-market value. If the weights are nondecreasing, they are coherent spectral weights in the loss convention (Acerbi, 2002). This endpoint observation does not establish ordering, monotonicity, or a single peak for arbitrary paths of spectra.

I.2 The scale of the residual ambiguity bound

For fixed $u_0 > 0$ and $\tau < u_0 p$, the baseline loading is $1 - p$. If a sequence of verified residual bounds satisfies $u_1(\tau)/\tau \to h \in [0, \infty]$, the common-regime formula yields

$$\lim_{\tau \downarrow 0} V(1-\tau) = G\left(\lambda \min\{1-p, hp(1-p)\}\right) - G\left(\lambda(1-p)\right), \tag{100}$$

where the minimum is $1 - p$ when $h = \infty$. A positive limiting gain requires $h < 1/p$ and active demand at the improved quote. If $\lambda(1-p) < v$, these conditions reduce to $h < 1/p$. The exact positive-net-gain test is (33); it uses $h < a_F/[\lambda p(1-p)] < 1/p$ for a limiting gain strictly above F.

For the linear-demand economy, $G(a) = [v-a]_+^2/(2h_Q)$, where $h_Q = b + \kappa$. Under the all-active restriction, the funding threshold is explicit:

$$a_F = v - \sqrt{[v - \lambda(1-p)]^2 + 2h_Q F}.$$

For Figure 10, $p = 0.02$, $u_0 = 0.9$, $u_1 = 0.2$, $\lambda = 0.08$, $v = 0.12$, $b = 0.0012$, and $\kappa = 0.0006$. Coverage is bounded below by $[v - \lambda(1-p)]/(b + \kappa) \simeq 23.11$ in each market. The gain is exactly zero for $\tau \le 0.004$, equivalently $\alpha \ge 0.996$. These are model calculations, not estimates of AI probabilities or recommended confidence levels.

I.3 The global costly-control solution under linear demand

The inner problem in (35) separates the effort needed to leave the pricing plateau from the additional reduction that earns a lower premium. For any smooth demand in the maintained active region, its gross-benefit function has derivatives

$$\frac{d}{dy} G(Ae^{-y}) = aQ(a), \qquad \frac{d^2}{dy^2} G(Ae^{-y}) = aQ(a)[\epsilon(a) - 1], \qquad a = Ae^{-y}.$$

The same surcharge elasticity as in Proposition 5 governs curvature. Gross benefit need not be concave in effort; a stationary solution must be compared with zero effort.

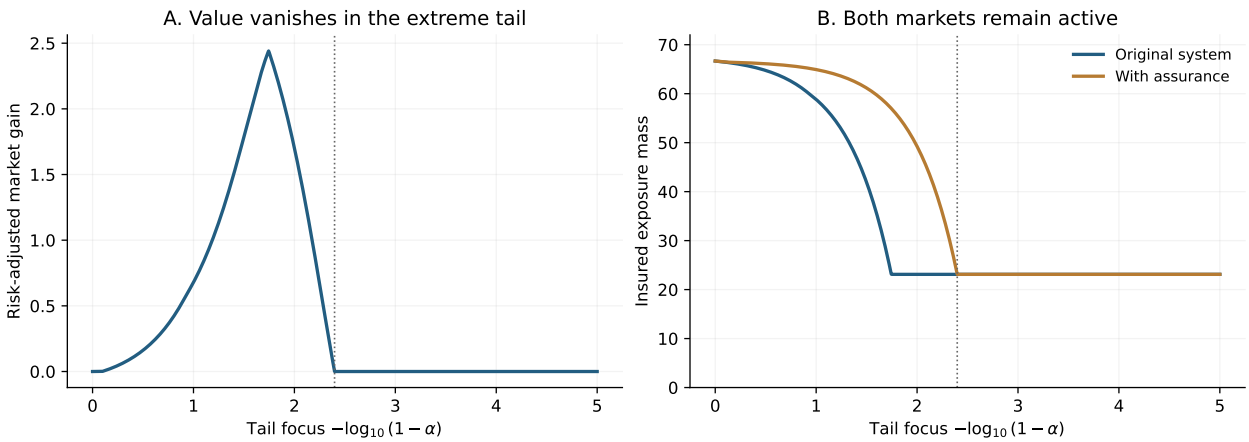


Figure 10: A fixed dependence control can lose its market value without market exit. The dotted line marks $1-\alpha = u_1 p$, beyond which both charges equal one. Panel A shows gross market gain; Panel B shows positive coverage throughout. Inputs are constructed.

With $h_Q = b + \kappa$ and $G(a) = (v-a)^2/(2h_Q)$ on $[0, A]$, write

$$f(y) = \frac{(v - Ae^{-y})^2 - (v-A)^2}{2h_Q} - \chi y.$$

Then $f'(y) = a(v-a)/h_Q - \chi$. If $\chi \geq v^2/(4h_Q)$, f is nonincreasing and $J = 0$. If $\chi < v^2/(4h_Q)$, define

$$a_- = \frac{v - \sqrt{v^2 - 4h_Q\chi}}{2}, \qquad a_+ = \frac{v + \sqrt{v^2 - 4h_Q\chi}}{2}.$$

If $A \leq a_-$, again $J = 0$. If $A > a_-$, the only possible positive global maximizer is $y_- = \log(A/a_-)$ and

$$J = \max\{0, f(y_-)\}. \tag{101}$$

Indeed, a decreases with y. The crossing at a_- is a local maximum, whereas a feasible crossing at a_+ is a local minimum. Together with $f(y) \to -\infty$, these facts exhaust the global candidates. When $f(y_-) > 0$, y_- is the unique inner maximizer. When its value is nonpositive, the outer problem has no positive optimal effort for $z > 0$. In particular, a positive stationary effort can exist yet lose to no investment.

For Figure 2, use $p = 0.02$, $u_0 = 0.9$, $\lambda = 0.08$, $v = 0.12$, $h_Q = 0.0018$, $\beta = 1$, and $c = 0.4$. The calculation gives $A = 0.0784$, $\chi = 0.4$, $J \simeq 2.10181$, and $y_- \simeq 2.51583$. The abandonment threshold is $z_* = J/\chi \simeq 5.25452$, or $\tau_* \simeq 9.40289 \times 10^{-5}$. At that threshold zero effort and $e \simeq 7.77036$ tie. Coverage without control is approximately 23.11; it is approximately 63.15 along the positive branch. These quantities and the corresponding confidence level are synthetic, not estimates or policy recommendations.

The exponential example also distinguishes uncertainty from actual risk. On a fixed positive branch, $u_0 e^{-\beta e^*} = (\tau/p)e^{-y^*}$ is independent of u_0. Increasing the admitted upper bound therefore calls for additional physical control to attain the same certified residual bound. If the true initial common-regime probability is held fixed, its controlled value falls on this branch, even though the underwriting quote is unchanged. At abandonment, the true common risk returns to its no-control level. This conditional prediction requires that the same true parameter belong to both ambiguity sets.

The general abandonment proof uses compactness of the set of controls whose resource cost can possibly be covered by market gains. It is not an exchange of pointwise limits and optimization. The conclusion can fail if complete common-regime elimination is feasible at finite cost below $B_{\max}$, if cost is not coercive, or if safety generates benefits outside the modeled insurance market. Even complete common-regime removal leaves finite-pool variation, as the next subsection shows.

The costly-control audit compares the closed-form inner maximum with direct mesh-and-refinement optimization in 400 linear-demand economies. It checks 1,600 global outer choices, including threshold ties, and 300 further choices in 100 nonlinear-demand economies. The maximum value discrepancy is below 7×10^{-15}; effort discrepancy on strictly profitable branches is below 3×10^{-7}. Thirteen feasible positive stationary candidates lose to zero in the inner problem, confirming the relevance of the global comparison. A separate 160-case ES dual linear-program audit has error below 2×10^{-16}. Another 150 constructed technologies verify the affordable-effort saturation bound, and 100 finite-elimination examples verify the countercase. The numerical searches do not replace the analytical global argument.

I.4 Pool size and tail confidence cannot be interchanged

Let $Y_{n,u}$ be the average of n unit Bernoulli claims in the common-regime family, and set

$$r_n(u, \alpha) = \sup_{0 \le \theta \le u} \mathrm{ES}_\alpha(Y_{n,\theta}).$$

The independent average is below a Bernoulli variable in convex order, so the supremum occurs at $\theta = u$. Fix $u_0 > 0$ and consider complete removal of the common regime, $u_1 = 0$. Define, with the same fixed demand and capacity function G,

$$V_n(\alpha) = G\big(\lambda[r_n(0, \alpha) - p]\big) - G\big(\lambda[r_n(u_0, \alpha) - p]\big).$$

Here n measures granularity within a unit of exposure; it is distinct from the endogenous mass of insurance Q.

At every finite n, the independent pool has a maximum payout of one with probability p^n. The original pool has that payout with probability $u_0 p + (1 - u_0)p^n \ge p^n$. Consequently,

$$V_n(\alpha) = 0 \quad \text{whenever} \quad 1 - \alpha \le p^n.$$

At every fixed $\alpha < 1$, Proposition 14 gives $r_n(0, \alpha) \to p$ and $r_n(u_0, \alpha) \to r(u_0, \alpha)$. Continuity of G therefore implies

$$\lim_{n\to\infty} \lim_{\alpha\uparrow 1} V_n(\alpha) = 0, \qquad \lim_{\alpha\uparrow 1} \lim_{n\to\infty} V_n(\alpha) = G(0) - G\big(\lambda(1 - p)\big) > 0. \tag{102}$$

The strict inequality uses $v > 0$, $p < 1$, and $\lambda > 0$. Thus the existing finite-pool approximation is not uniform as $\alpha \uparrow 1$; its factor $1/(1 - \alpha)$ cannot be ignored. If a fixed positive common-regime bound remains after the project, both iterated limits are instead zero. The noncommutation in (102) concerns complete removal of the common component.

The numerical program evaluates finite binomial laws directly and confirms their exact saturation thresholds. An extremely small probability need not be empirically relevant at a chosen policy horizon.

The tail-focus audit checks 250 finite-law ES linear programs (error below 3×10^{-16}), 2,880 convex-order comparisons, 250 robust mixture grids, 1,000 cost-coverage comparisons, 150 scaling limits, 120 binomial saturation cases, and 100 spectral endpoints. These synthetic checks support the formulas, not empirical relevance or the universal proof.

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